Clientelism, Elite Capture and Structural Violence: Theory and Structural Estimation (Job Market Paper)

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Abstract

This paper links electoral competition to structural violence. I extend a standard model of electoral competition with political clientelism and elite capture, allowing voters to choose how they vote and parties to choose how much clientelistic pressure to exert. In equilibrium, clientelism worsens inequality and reinforces elite capture, in contrast to earlier work in which clientelism favours the poor. I define structural violence as the welfare gap between this equilibrium and an egalitarian benchmark. To measure it, I structurally estimate the clientelism side of the model using household survey data from rural West Bengal, where Gram Panchayat elections are contested on party lines, The estimates show that parties favour elite households in allocating private transfers. Making voter behaviour and clientelistic pressure endogenous strengthens this result: the welfare weight on elite is 71 to 87 percent higher than on non-elites. A counterfactual compares welfare under clientelism.

Keywords: Clientelism, Elite Capture, Private goods, Electoral Competition, Structural Violence

JEL classification: D02, D72, H42

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1 Introduction

Discussions on the political economy of development point to decentralisation as a method of governance because of its positive impact on empowerment, autonomy, transparency, accountability, and its ability to efficiently identify people's preferences and tailor location specific policies. However, it comes with its own problems.

This paper is about the inefficiencies that arise from decentralisation, focusing on its susceptibility to political clientelism and elite capture. The paper incorporates these dynamics into a formal model of electoral competition based on the theoretical model by Bardhan and Mookherjee (2023). The analysis shows how democratic processes in unequal societies can reinforce, and in some cases deepen, the existing patterns of inequality. This paper contributes to the literature by linking electoral competition and inequality to what Galtung (1969) conceptualises as "Structural Violence".

The central question this paper addresses is how local elites capture political parties in a decentralised economy where the majority of the population is poor. Once captured, political parties relying on clientelism design policies that allocate only minimal transfers to the poor while disproportionately favouring the rich, yet still manage to secure sufficient vote shares to remain in power. This raises a fundamental question: What are the implications for inequality in a democracy shaped by the joint influence of clientelism and elite capture?

The literature on political clientelism defines it as a phenomenon that occurs when political parties condition transfers of goods to voters in exchange for support. This type of targeting favours poor voting groups or higher loyalty groups and groups that are more likely to reciprocate with their votes. In addition, marginal voting groups are also strategically targeted Stokes (2005); Kitschelt and Wilkinson (2007); Keefer and Khemani (2009); Dunning and Nilekani (2013); Bardhan and Mookherjee (2020). It causes politicians to favour short-term recurring private transfer programs such as public employment, loans and subsidies at the expense of the public good or private benefit of a long-run nature such as education or health services. While clientelism has been hailed for its redistributive impact and its role in addressing gaps in state provided social services, it is also widely regarded as having a detrimental impact on both democracy and development.

Another major inefficiency is elite capture, a form of corruption in which wealthy special interest groups influence political parties through campaign donations or bribes, inducing them to adopt

policies that favour the rich, such as preferential treatment, tax benefits at the expense of the poor. Elite capture has long been central to the debate on the disadvantages of democratic decentralisation Bardhan and Mookherjee (1998, 2000, 2002); Bardhan (2002).

Previously, the literature on elite capture and clientelism developed largely in parallel, without any overlap. Recent work by Bardhan and Mookherjee (2023), demonstrates that clientelism and elite capture coexist.

This paper extends the electoral competition model developed by Grossman and Helpman (1996); Bardhan and Mookherjee (2023), in which voters have different types, and interests groups can influence political parties' policy platforms. The proposed framework shows complementarity between clientelism and elite capture through two key extensions. First, the model endogenises the voter's choice of types based on initial endowment. Second, the model incorporates an endogenous clientelistic control parameter, allowing political parties to tailor their engagement with voter groups to their economic status. These extensions offer a more refined understanding of how economic inequality shapes voter behaviour and party strategy.

My model focuses on the problem of allocating a government benefit comprising a single private good (such as food, fuel, housing ...) across two groups (Elite and Non-Elite) with varying initial endowments. Theoretical literature provides evidence that poor voting population prefers private good transfer over public goods or programmatic policy offerings because of their risk averse and present biased tendencies Bardhan and Mookherjee (2018), the empirical side of the literature confirms this claim Dixit and Londregan (1996); Brusco et al. (2004); Calvo and Murillo (2004); de Janvry et al. (2014); Castañeda Dower and Pfutze (2015); Holland (2016); Lo Bue et al. (2021).

Elite capture involves the elite group's contribution to the campaign funds of two political parties in exchange for a proposed allocation of the government benefit favoured by this group. Probabilistic monitoring of the voting behaviour of each group by each party represents clientelism, followed by selective delivery of benefits conditioned on this information. Political parties are modelled in a Downsian fashion. Each party selects an electoral strategy comprising a proposed allocation of benefits, and a level of campaign finance which is used to 'persuade' voters along with an endogenised function of clientelistic control parameters. Citizen's voting decisions are based partly on pre-existing loyalties and campaign advertising, partly on strategic considerations and an endogenised choice function that determines the type of voter based on the initial endowment.

To derive clear predictions, I assume that an electoral contest leads to a unique equilibrium characterised by the probabilities of either party winning and the corresponding allocation of benefits. This model demonstrates that political parties assign a higher welfare weight to wealthier groups, which in turn contributes to a rise in overall inequality. This political process distributes endowments on the basis of political characteristics, which are in general different from the economic characteristics.

Structural violence as defined in sociology " is a form of violence where some social structure or social institution may harm people by preventing then from meeting their basic needs or right" Galtung (1969). Following this idea, I define structural violence as the gap between the welfare the poor actually receive and the welfare they receive under the most egalitarian allocation. In the equilibrium of the model, the transfers of the private good to each voting group leaves the rich with higher ex-post consumption than the poor. Under the benchmark of the "perfect democracy", in which both clientelism and elite capture are absent, equilibrium transfers instead equalise consumption across groups. Structural violence is the difference in ex-post welfare of the poor between the distorted equilibrium and this benchmark.

I take the model to household survey data from West Bengal, a state in eastern India. The clientelistic parameters are estimated structurally from the first order conditions and equilibrium equations derived in the theory section. Although the model allows for elite capture, the data contain no information on campaign finance or elite capture with which the capture parameter could be identified, so the estimation is confined to clientelism. I use elections to the gram panchayat (GP), the lowest tier of elected government in India, and assume that households receive clientelistic transfers in the form of private benefits from the party than controls the GP after the elections.

I estimate two specifications. In the first, the parameter governing voters' choice type is set exogenously, and the clientelistic parameter measures the clientelism experienced by households, distinguished by elite status and by whether the Left or Non-Left party won the GP. In the second, both the voting choice and clientelistic pressure are endogenous, and the clientelistic parameter is interpreted as the pressure a party exerts on each group. The structural approach allows me to quantify clientelistic pressure at different levels of electoral competition and to recover the political welfare weights that determine how parties value the welfare of each group once in office.

The first specification shows that all households experience clientelism, and that its severity

falls as electoral competition weakens. In GPs controlled by Non-Left parties, the distribution of private goods in regressive, and it holds for every value of the voter type parameter and at every level of electoral competition. In GPs controlled by the left, the distribution can instead be progressive or egalitarian, depending on the value assumed for the voter type parameter.

The second specification removes this ambiguity. Once voters choose their type, estimated clientelistic pressure is positive for non-elites and zero for elites in both sets of GPs. Elites optimally vote non-instrumentally, so no amount of pressure changes their vote; the party's best response is to exert none and to compete for elite votes through private transfers instead. The resulting distributive preferences are strictly regressive under both parties. the welfare weight on the exceeds that on non-elites by 71 percent in the Left GPs and by 86.77 per cent in Non-Left GPs. These preferences are robust to changes in the level of political competition and closely match those implied by the point identified estimates.

Overall, the second specifications supports the central prediction of theory: clientelism operates regressively and worsens inequlity. This reverses the result of Bardhan and Mookherjee (2023), in which clientelism makes allocations more progressive and only elite capture is regressive. The reversal arises because voter type and clientelistic control are endogenous here, once elites choose to vote non-instrumentally, parties direct pressure at the poor and transfers at the rich.

The remainder of this paper is as follows. The next section discusses the relationship to several other literature. Section 3 develops and extends the theoretical framework along with welfare implications. Section 4 establishes the link between the model and structural violence, via the transmission channel of systematic inequality generated through clientelism and elite capture. Finally, Section 5 concludes.

2 Related literature

There is a vast theoretical as well as empirical literature on clientelism and elite capture. In the discussion that follows, I will restrict the scope to those contributions most directly relevant to the arguments in this paper.

The literature on political clientelism and democracy is divided into two strands: one examining its relationship with democracy, and the other exploring its connection with development. The first approach used to study the relation between clientelism and democracy assumes that clientelism was bound to disappear as democracies progressed Gellner et al. (1977); Landé (1977);

Scott (1977); Schmidt (1977). A second approach focuses on understanding how clientelism adapts to changing circumstances Chubb (1982); Eisenstadt and Roniger (1984); Kettering (1986); Reynolds (1986).

An ample body of literature provides evidence that, for political parties, targeting poor voters with clientelistic practices yields higher expected returns, owing to their risk-averse and present-biased preferences Scott (1977); Dixit and Londregan (1996); Kitschelt (2000); Calvo and Murillo (2004); Stokes (2005); Keefer and Vlaicu (2007); Lyne (2007); Robinson and Verdier (2013).

Economic development compels political parties to seek alternatives to clientelism, as rising incomes and diminishing utility of income increase the cost of sustaining clientelistic strategies. Development reduces the probability of political parties securing power through clientelistic practices. As it generates negative externalities, it creates incentives for parties to maintain conditions of underdevelopment, as these conditions preserve the effectiveness of clientelistic strategies in mobilising electoral support Kitschelt and Wilkinson (2007); Robinson and Verdier (2013); Hiken (2007); Schaffer (2007).

Moreover, changes in income are shown to influence shifts in voter preferences Kitschelt and Wilkinson (2007); Hicken (2011). Empirical literature on clientelism across developing countries presents a negative correlation between development and clientelism as shown by Wantchekon (2003); Brusco et al. (2004); Khemani (2015); Bardhan and Mookherjee (2018).

Duflo et al. (2005) argues that, in decentralised systems of government, may more efficiently identify people's preferences than centralised systems, but they may be more vulnerable to capture. Bardhan and Mookherjee (2000) emphasises inequality, poverty, political awareness, and electoral dynamics as the principal determinants of elite capture.

Theoretical models in political economy that examine the role of historical conditions in shaping long-run development similarly rely on capture-based distortions to explain persistent institutional and economic inequalities Acemoglu et al. (2001); Acemoglu and Robinson (2008); Banerjee and Iyer (2005).

A substantial body of empirical literature has explored various approaches to measuring elite capture and assessing its impact on the allocation and distribution of government expenditure Bardhan and Mookherjee. Dilip (2006); Bardhan and Mookherjee (2006); Acemoglu et al. (2014). In the worst case, elite capture may lead to obstruction in adoption of new technology Acemoglu and Robinson (2000). Palaniswamy and Krishnan (2012) shows the adaptive nature

of elite capture to preventative policy measures. Besley and Burgess (2002) shows that elite capture can be avoided if there is a strong and free media.

This paper is a modification of the already existing long literature on probabilistic voting and electoral competition. Standard probabilistic models of voting and electoral competition in the Downsian tradition have been developed by Lindbeck and Weibull (1987); Dixit and Londregan (1995). These models provide determinants of political success of special interests; however, these frameworks do not account for the possibility of interest groups capturing the policy platforms of political parties. Building on this limitation, Grossman and Helpman (1996) develop a model where interest groups contribute to campaign funds in exchange for favourable policies. Austen-Smith (1987); Baron (1994) address similar issues of capture and voting behaviour. In both models, interest groups take the policy platform as given; however, Grossman's model allows for interest groups to influence the parties' policy platform.

Bardhan and Mookherjee (2023) extends on Grossman's model and shows that elite capture and clientelism coexist. While the model treats voter types and the party's clientelistic control as exogenously given, this paper endogenises both. Specifically, voters are allowed to choose whether to vote instrumentally or non-instrumentally, and political parties can strategically determine the level of clientelistic pressure based on the voters' initial endowments. The inequality caused as a result helps explain the systemic violence on the poor.

Empirical literature on clientelism in India and other developing countries is largely reduced form. Using the household surveys from Rural West Bengal Bardhan et al. (2009, 2024) show that voters reward incumbents for private welfare benefits but not for local public goods, a pattern they interpret as evidence of clientelism. Related work documents that partisan alignment and party loyalty shape the allocation of welfare programmes and that incumbents gain votes as a result Anderson et al. (2015); Shenoy and Zimmermann (2026); Dey and Sen (2016). Similar electoral returns to targeted transfers are found in other countries Wantchekon (2003); Manacorda et al. (2011); Labonne (2013); De La O (2013). These studies establish that benefits buy political support, but the estimated vote responses combine several distinct forces: the value voters place on the benefit itself, the fear of losing it it they do not support the provider, and their beliefs about which party will win.

Separating these forces required estimating the primitives of a voting model. A body of work estimates probabilistic voting and electoral competition models structurally Coate and Conlin (2004); Strömberg (2008); Kawai and Watanabe (2013); Kendall et al. (2015), but application

to developing countries remain rare. Finan and Schechter (2012); Finan and Mazzocco (2021) 2021 estimate how brazilian legislator allocate public funds under electoral incentives, and Ujhelyi et al. (2021) estimate a model of protest voting in India. The closest paper to mine is Cruz et al. (2024), who estimate a model of voter choice in Philippine Mayoral elections in which vote buying enters alongside policy and candidate valence. In their framework vote buying shifts voter utility directly. In the model estimated here, clientelism instead operates through the threat that a winning party withholds benefits from a non-supporters, so its effect on votes depend on the each party's expected probability of winning.

This paper also relates to work that combines structural models with quasi-experimental evidence in development economics Todd and Wolpin (2023); Attanasio et al. (2012). Mookherjee and Nath (2023) use the same West Bengal data to simulate the effect of replacing discretionary local budgets with formula based grants, without estimating the underlying political parameters.

I contribute to this literature by structurally estimating the clientelism mechanism of the model developed in this paper, using the household benefit and voting data of Bardhan et al. (2024). I estimate two specifications, first, the parameters governing clientelistic pressure and voter choice is taken as exogenous, closely following formulation in Bardhan and Mookherjee (2023). Second, the parameters governing clientelistic pressure and voter choice is endogenised. Although the theoretical model model allows for elite capture, the estimation abstains from it, because the data contains no information on campaign finance or elite contributions with which the capture parameter could be identified. The estimates should therefore be read as charactering clientelism conditional on this restriction.

3 Model

Consider a village with multiple voters belonging to several households. These households are clubbed into two groups based on their endowment of a single consumption good or asset. These consumer goods or assets can be supplemented by the local government in terms of direct transfers. Analysis by Bardhan and Mookherjee (2018) that politicians have low incentives to provide public good and voters prefer private transfers over public goods. Elites are members of a group characterised by a higher ex-ante endowment, whereas non-elites belong to a group with a lower ex-ante endowment. The ex-ante endowment is denoted by ω and is public information across all groups and political parties. Fraction of households in each group is given by μ_i. The

fraction of elite households (μ_e) is less than fraction of non-elite households (μ_c). Households with endowment are given by ω_i, where $i = \{e, c\}$ and $\omega_c < \omega_e$. The local government can augment the endowment of a group i households with a transfer $q_i \geq 0$, subject to the budget constraint:

$$\sum_{i \in \{e,c\}} \mu_i q_i \leq B$$

$$\mu_c q_c + \mu_e q_e \leq B$$

Household i ends up with utility $u(\omega_i + q_i)$, where u is strictly increasing, differentiable and strictly concave.

3.0.1 Optimal Allocation

The utilitarian optimal allocation maximises:

$$\sum_{i \in \{e,c\}} \mu_i u(\omega_i + q_i) \tag{1}$$

subject to budget constraint and non-negativity constraints. This allocation is maximally progressive: if the program scale B is sufficiently large, it achieves full consumption equalisation across all groups. Otherwise, benefits are exclusively allocated to the poorest groups ($as\ , \omega_c < \omega_e$), ensuring consumption equality ($\omega_c + q_c^* = \omega_e + q_e^*$) within these groups. [1].

3.0.2 Quasi-Utilitarian Allocation Maximiser

The Quasi-Utilitarian objective with 'political' welfare weight δ_i for group i replaces the objective function:

$$\sum_{i \in \{e,c\}} \mu_i \delta_i u(\omega_i + q_i) \tag{2}$$

The quasi-utilitarian optimal allocation discriminates in favour of groups with higher welfare weights, while equalising consumption within each group. It can involve regressive targeting if welfare weights are rising steeply enough with wealth. In particular, an elite group may end up receiving some of the benefit if the group has a sufficiently large political welfare weight, even though they may not be in the utilitarian optimum.

[1] Optimisation problem solved in Appendix A.1

Political economy considerations, which distort allocations across the different groups in equilibria of the model, will be mirrored by deviations of these political welfare weights from unity. The rest of this section is based on deriving the equilibrium political welfare weight for group i, along with examining the welfare implications as a result of different welfare weights across different groups. The quasi-utilitarian optimal allocation discriminates in favour of groups with higher welfare weights. Here, elites may end up with some welfare weight, even though they may not be in the utilitarian optimum. [2].

3.1 Elections

There are two parties, denoted L and R, modelled in the Downsian fashion - pursuing power for its own sake, without any personal or ideological preferences over policy. Each party seeks to maximise its chances of winning by increasing patronage to secure votes while simultaneously enhancing elite appeasement. The policy of party p is denoted by electoral platform consisting of feasible allocation of transfers $\{q_i^p\}$ satisfying: $\sum_{i\in\{c,e\}} \mu_i q_i^p \leq B;\ q_i^p \geq 0$. Π is the set of all such feasible policies.

3.2 Voting

Voting occurs during elections, with voters from elite and non-elite groups casting their ballots. Voters know about their own ex-ante utility and the transfers $\{q_i^p\}$ offered by the political party. Voting behaviour is decided under a number of influences.

3.2.1 Loyalties

Each voter possesses an ex-ante loyalty to one of the two parties (L or R), shaped by historical factors. They may also place importance on candidate characteristics such as gender, caste, and personal reputation.

A voter's ex-ante propensity ϵ_i to vote for party $L's$ candidate is drawn from a uniform distribution centred at l_i over the range of width $\frac{1}{\sigma_i}$. Here l_i is the average loyalty, which depends on historical, social, ethnic, gender, or ideological-based affiliation to the party. While each group has an average loyalty l_i individual voters may have a personal variation in their preferences. The above distribution is modelled by adding a uniform random deviation given by equation 3

[2]Maximisation done in Appendix A.2

and is shown in the Figure 1:

$$\epsilon_i \sim U(\frac{-1}{2\sigma_i}, \frac{1}{2\sigma_i}) \tag{3}$$

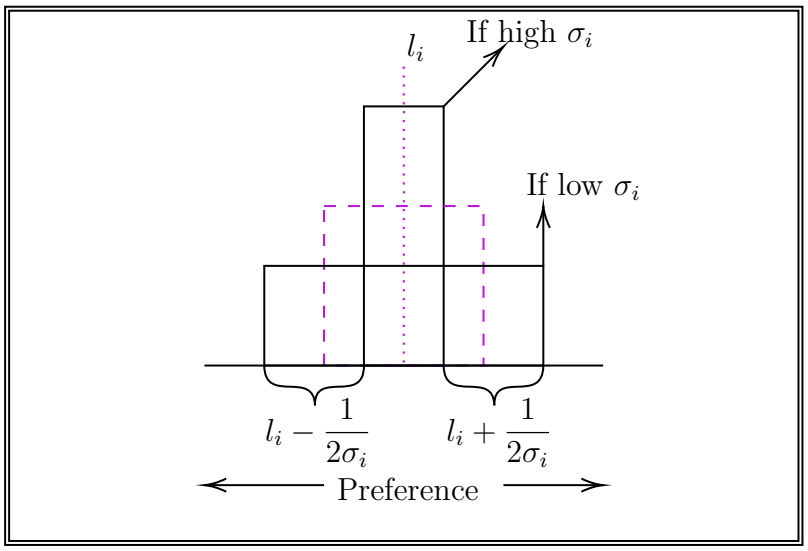


Figure 1: Loyalty distribution

σ_i is the swing parameter. Higher swing parameter means the voters are more loyal, the variation is centred around l_i. Lower swing parameter means the voters are less loyal, the variation is dispersed. Initial preference to vote for party L is then given by:

$$l_i + \epsilon_i > 0 \tag{4}$$

3.2.2 Non-Instrumental Voting

Households are presumed to vote based on a comparison of their expected utility under policies espoused by the two parties. This group of voters may also be characterised as aware and engaged, reflecting their informed decision making active participation in the electoral process. The expected utility of a voter from voting party P is given by:

$$N_i^P \equiv u(\omega_i + q_i^P) - u(\omega_i)$$

A voter of this type will be more inclined to vote for L when $N_i^L - N_i^R > 0$:

$$N_i^L - N_i^R \equiv u(\omega_i + q_i^L) - u(\omega_i + q_i^R) \tag{5}$$

3.2.3 Instrumental Voting

Parties have the capacity to condition the delivery of transfers to any households depending on how it votes. Party p can withhold transfers to group i, if party p gets in power and group i has not voted for them. In the absence of elite capture, party p can have clientelistic control over both elite and citizen groups, since the elites lack the financial leverage over the political party. z_i^p is a parameter representing the strength of the party p's clientelistic control over voters in group i.

My model differs from the original framework by endogenising clientelistic control, allowing the political party to determine its extent based on the initial endowment of the voting group. The clientelistic parameter is now endogenised as follows:

$$z_i^p(\omega_i) = (0, 1)$$

Here, the clientelistic control of the political parties is a function of the initial endowment of the voting group. (The political party chooses its clientelistic control parameter based on the initial endowment of the voting population.) The function gives an output of $(0, 1)$. An output of 0 means that a political party has a low control over the voting group, and an output of 1 means a political party has a high control over the voting population.

For simplicity, I am considering $z_i^p(\omega_i) \equiv z_i^p$. A voter in group i obtains the expected utility by voting for L, given by:

$$I_i^L \equiv \gamma^L u(\omega_i + q_i^L) + (1 - \gamma^L) \left[z_i^R u(\omega_i) + (1 - z_i^R) u(\omega_i + q_i^R) \right]$$

Here γ^L is the voters probability assessment that L will win the elections. The corresponding expected utility of voting R:

$$I_i^R \equiv \gamma^L \left[z_i^L u(\omega_i) + (1 - z_i^L) u(\omega_i + q_i^L) \right] + (1 - \gamma^L) u(\omega_i + q_i^R)$$

A voter of this type will be more inclined to vote for L if $I_i^L - I_i^R > 0$

$$I_i^L - I_i^R \equiv \gamma^L z_i^L [u(\omega_i + q_i^L) - u(\omega_i)] - (1 - \gamma^L) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)] \tag{6}$$

3.3 Voting Behaviour

Each household votes as a result of the addition of all the above considerations from equations (4), (5), (6). A population's policy based voting behaviour blends non-instrumental and instrumental voting motives with an endogenised choice function added to the effects of non-policy based loyalty. The endogenised choice function is given by:

$$\Theta_i \equiv \ \theta(\omega_i) \in [0,1]$$

Voters are now able to endogenously determine their own type based on their initial endowment. A voter's behaviour is given by:

$$\Theta_i[N_i^L - N_i^R] + (1-\Theta_i)[I_i^L - I_i^R] + \epsilon_i > 0 \tag{7}$$

The resulting vote share of L will be [3]:

$$\begin{aligned} S_L(\gamma^L; q^L, q^R) \equiv \frac{1}{2} + \sum_i \mu_i l_i \sigma_i + \sum_i \mu_i \sigma_i \{\Theta_i \left[u(\omega_i + q_i^L) - u(\omega_i + q_i^R)\right] \\ + (1-\Theta_i)(\gamma^L z_i^L \left[u(\omega_i + q_i^L) - u(\omega_i)\right] - (1-\gamma^L) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)]\} \end{aligned} \tag{8}$$

Here the equation 8 is the resulting vote share of party L, where γ^L is the voters probability assessment that L will win the elections. Instrumental motive will depend on voters assessment of γ^L of election outcome. *The equilibrium probability of winning is determined by a fixed point condition in which the probability of electoral success coincides with the self fulfilling belief about winning. The equilibrium probability is bounded away from both extremes. It has a lower bound strictly above 0.5 and and upper bound strictly below 1. The equilibrium relationship is concave over the interval from 0.5 to 1.*

3.3.1 Winning Probability and the Equilibrium Fixed Point

S_L affects but does not determine election outcome. χ is the aggregate uncertainty represented by vote counting error, randomness in voter turnout. Therefore, party L wins if:

$$S_L + \chi > 0$$

[3] Vote share will be calculated in Appendix A.3

This induces a smooth monotone relationship between vote share and winning probability. Define:

$$\phi : S_L \to [0, 1]$$

The function $\phi(S_L)$ is assumed to be strictly increasing and smooth, satisfying:

$$\phi(0) = 0, \quad \phi(1/2) = 1/2, \quad \phi(1) = 1$$

When the vote share equals half, the probability of winning is also half. Further,it is assumed that ϕ is convex $[0, 1/2]$ and concave $[1/2, 1]$, consistent with figure 2.

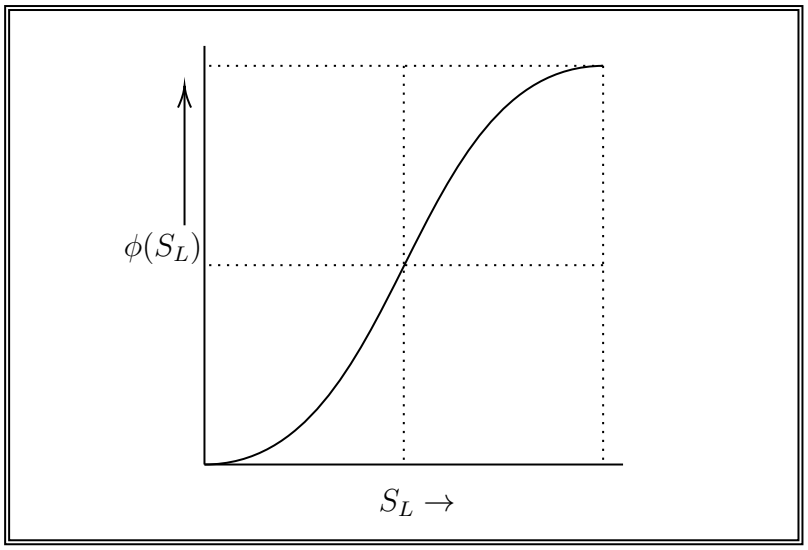


Figure 2: $\phi : S_L \to [0, 1]$

The fixed point equation

The model is closed by requiring voters' belief about the probability of party L winning with probability γ^L, this enters the information term of the vote share $S_L(\gamma^L)$, which in turn determines the winning probability $\phi(S_L(\gamma^L))$. The equilibrium winning probability $\gamma_L^* \equiv \gamma(q^{L*}, q^{R*})$ is defined as the fixed point mapping $\phi(S_L(\cdot; q^L, q^R))$:

$$\gamma_L^* = \phi(S_L(\gamma_L^*; q^L, q^R)) \tag{9}$$

γ_L^* is the voters equilibrium probability assessment that L will win the elections. For this fixed point to exists and be unique, the mapping $\gamma^L \to \phi(S_L(\gamma^L))$ must be a strict contraction.

The equilibrium fixed point γ^* lies between 0.5 and 1. With vote share having an upper bound $\bar{S}_L$. Party L will win election if and only if its equilibrium probability of winning is $\gamma^* > 1/2$

The Upper bound of Vote Share

Let $\bar{S}_L$ be the highest vote share party L can achieve, attained when probability of winning is $\gamma^L \to 1$:

$$\bar{S}_L \equiv \lim_{\gamma^L \to 1} S_L(\gamma^L)$$

As $\gamma^L \to 1$, the probability of winning of the opponent becomes $(1-\gamma^L) \to 0$, so the information term collapses to L's campaign alone. The vote share becomes:

$$\bar{S}_L = \frac{1}{2} + \sum_i \mu_i \sigma_i l_i + \sum_i \mu_i \sigma_i \{\Theta_i [N_i^L - N_i^R] + (1 - \Theta_i) z_i^L [u(\omega_i + q_i^L) - u(\omega_i)]\}$$

This is the tightest point at which the contraction condition must hold, if the mapping is a contraction $\bar{S}_L$, it is a contraction everywhere on $[1/2, \bar{S}_L]$.

Since the function ϕ is convex on $[0, 1/2]$ and concave on $[1/2, 1]$, It has an inflection point at $S_L = 1/2$. The slope of this function $\phi'(S_L)$ in increasing in the convex region and is decreasing on this interval. Therefore the slope is maximum at $S_L = 1/2$ and minimum at $\bar{S}_L$. The relevant slope for the contraction condition, the one that binds at the upper bound of vote share is: $\phi'(\bar{S}_L)$

Differentiating equation 8 with respect to γ^L:

$$\frac{\partial S_L}{\partial \gamma^L} = \sum_{i \in \{e,c\}} \mu_i \sigma_i (1 - \Theta_i)(z_i^L [u(\omega_i + q_i^L) - u(\omega_i)] + z_i^R [u(\omega_i + q_i^R) - u(\omega_i)])$$

Applying the chain rule to the winning probability, The derivative of the full mapping $\gamma^L \to \phi(S_L(\gamma^L))$ is:

$$\frac{\partial\ \phi(S_L(\gamma^L))}{\partial\ \gamma^L} = \phi'(S_L) \frac{\partial S_L}{\partial \gamma^L}$$

For unique fixed point, the mapping must satisfy:

$$\frac{\partial\ \phi(S_L(\gamma^L))}{\partial\ \gamma^L}|_{S_L = \bar{S}_L} < 1$$

applying the chain rule yields

$$\phi'(S_L)\frac{\partial S_L}{\partial \gamma^L} < 1$$

This is the contraction condition evaluated at its tightest point. Rearranging:

$$\phi'(\bar{S}_L) < \frac{1}{\sum_{i\in\{e,c\}} \mu_i\sigma_i(1-\Theta_i)(z_i^L[u(\omega_i+q_i^L)-u(\omega_i)]+z_i^R[u(\omega_i+q_i^R)-u(\omega_i)])}$$

Hence, a unique fixed point exists whenever the responsiveness of the function, $(\phi'(\bar{S}_L))$, is sufficiently small relative to the marginal effect of the vote share function.

Since ϕ is concave on $[1/2, 1]$, ϕ' is non-increasing, therefore $\phi'(t) \leq \phi'(1/2)\ \forall\ \ t \in [1/2, \bar{S}_L]$. If $\bar{S}_L \geq 1/2$, then

$$\phi(\bar{S}_L) = \phi\left(\frac{1}{2}\right) + \int_{1/2}^{\bar{S}_L} \phi'(t)\ dt \leq \frac{1}{2} + \phi'\left(\frac{1}{2}\right)\left(\bar{S}_L - \frac{1}{2}\right).$$

Hence $\gamma_L^* \leq \min\left\{\phi(\bar{S}_L), \frac{1}{2} + \phi'(\frac{1}{2})(\bar{S}_L - \frac{1}{2})\right\} < 1$, therefore $\gamma_L^* \in (1/2, 1)$.

The condition $\phi(\bar{S}_L)' \cdot \partial S_L/\partial\gamma^L < 1$ is the contraction mapping condition, evaluated at $\bar{S}_L$ the vote share achieved when party L maximises its information campaign intensity. Since party L's influence over the belief-to-outcome mapping is strongest at this point, imposing the contraction condition here guarantees it hold everywhere on $[1/2, \bar{S}_L]$, ensuring a unique fixed point γ_L^* exists. The fixed point is self-fulfilling equilibrium belief about L's winning probability.

In the context of democratic elections, it is logically and institutionally implausible for any political party to possess a probability of winning equal to one. Democratic systems are inherently characterised by competition, uncertainty, and the possibility of electoral turnover. A probability of one would imply an absolute certainty of electoral victory, effectively eliminating the role of voter choice and undermining the fundamental principles of democratic governance. Such an outcome would more closely resemble an authoritarian regime, where political competition is either severely constrained or entirely absent. Therefore, any credible model of democratic elections must account for the inherent uncertainty in electoral outcomes, ensuring that no party is guaranteed victory with absolute certainty.

The equilibrium probability party L winning is determined by the equilibrium vote share $\gamma_L^* \equiv \gamma(q^{L*}, q^{R*})$ is fixed point of the function $\phi(S_L(.; q^L, q^R))$:

The equilibrium vote share is then given by:

$$S_L(\gamma_L^*; q^L, q^R) \equiv \frac{1}{2} + \sum_i \mu_i l_i \sigma_i + \sum_i \mu_i \sigma_i \Big\{ \Theta_i \left[u(\omega_i + q_i^L) - u(\omega_i + q_i^R) \right] + (1 - \Theta_i)(\gamma_L^* \; z_i^L \; \left[u(\omega_i + q_i^L) - u(\omega_i) \right] - (1 - \gamma_L^*) \; z_i^R \; \left[u(\omega_i + q_i^R) - u(\omega_i) \right] \Big\} \tag{10}$$

Similar arguments applies to party R.

3.4 Pure Clientelism

In the previous section, we examined the quasi-utilitarian allocation equation that a political party seeks to maximises upon winning an election. The voter behaviour in this framework is modelled as the combination of policy-based preference and non-policy-based loyalty toward party p. The resulting vote share is characterised by an S-shaped function, ensuring that the equilibrium vote share for any party remains strictly less than one. Consequently, the equilibrium probability of a party p winning the election is endogenously determined by the aggregate equilibrium beliefs of voters regarding that party's probability of success.

We begin by considering a scenario characterised by the absence of elite capture and a reliance on campaign financing for vote accumulation. The political party's campaign policy is then taken as $\{q_i^P\}$. There is a unique equilibrium which is characterised as follows, In this equilibrium, party p selects transfers $\{q_i^p\}$ which maximises the quasi-utilitarian welfare function:[4]

$$\sum_{i \in \{e,c\}} \mu_i \sigma_i [\Theta_i + (1 - \Theta_i)\gamma_p^* \; z_i^p] u(\omega_i + q_i^p) \tag{11}$$

over the set of feasible allocations, taking γ_p^* as given, the equilibrium probability of party p winning.

The above results show that the political welfare weight is similar to that of Bardhan and Mookherjee (2023) with two important extensions. First, the model incorporates an endogenous choice function Θ_i, which enables voters to select their type based on their initial endowment. Second, it introduces an endogenous clientelistic control parameter, $z_i^p(\omega_i)$, allowing political parties to determine the degree of clientelistic engagement with the voting group as a function of that group's initial endowment. These additions provide a more nuanced understanding of how economic disparities influence both voter behaviour and party strategy. Furthermore,

[4]Equation for welfare weight and equation 12 is calculated in Appendix A.4

these extensions help in determining the welfare impact created as a result of different political welfare weights.

$$\delta_i = \sigma_i[\Theta_i + (1 - \Theta_i)\gamma_p^* z_i^p]$$

This implies that in equilibrium, party p chooses a benefit allocation which maximises a quasi-utilitarian welfare function which assigns a welfare weight of δ_i to group i taking γ_p^* as given.

Proposition 1. *If both political party's assign the same welfare weight to elite δ_e and non-elite δ_c, then equilibrium policy platform converges across parties, such that the final utility of elite and non-elite is invariant of electoral outcomes. Intergroup utility need not be equalized.*

This proposition highlights that electoral competition does not necessarily generate differences in policy outcomes when competing parties assign the same political welfare weights to elites and non-elites. In this case, electoral competition leads both parties to converge on the same policy platform, making the final utility of each group independent of which party wins the election. However, this convergence does not imply equality in welfare across groups. Differences in initial endowments, political influence, or other group specific characteristics may persist in equilibrium, so intergroup utility need not be equalized even when electoral competition produces identical policy platforms. The proof of this proposition is in Appendix A.6

Corollary 1. *Suppose the conditions of Proposition 1 hold, such that equilibrium policy platforms converge and $q_i^{L*} = q_i^{R*} = q_i^*$ for all groups $i \in \{e, c\}$. Then electoral outcomes are determined solely by voter loyalties, swing propensities, and the structure of clientelistic competition.*

Substituting the common equilibrium transfer vector into the voting equation yields:

$$\gamma_L \equiv \phi\Bigg(\frac{1}{2} + \sum_i \mu_i \sigma_i l_i + \sum_i \mu_i \sigma_i (1 - \Theta_i) \cdot \\ (\gamma_L \; z_i^L \; [u(\omega_i + q_i^*) - u(\omega_i)] - (1 - \gamma_L) \; z_i^R \; [u(\omega_i + q_i^*) - u(\omega_i)]\Bigg) \quad (12)$$

The equation 12 highlights three cases.

No Clientelism $z_i^p = 0$: In the absence of clientelistic pressure, electoral outcomes depend only on voter loyalties and swing propensities:

$$\gamma_L^* = \phi(1/2 + \sum_i \mu_i \sigma_i l_i)$$

Hence, if $\sum_i \mu_i \sigma_i l_i > 0$ party L enjoys a loyalty advantage and therefore $\gamma_L^* > 1/2$.

Symmetric Clientelism $z_i^L = z_i^R = z_i$: When both parties exert identical clientelistic pressure, the probability of victory becomes

$$\gamma_p \equiv \phi\left(\frac{1}{2} + \sum_i \mu_i \sigma_i l_i + \sum_i \mu_i \sigma_i (1 - \Theta_i) \cdot ((2\gamma_p - 1) z_i \ \ [u(\omega_i + q_i^*) - u(\omega_i)])\right) \tag{13}$$

In this case, clientelism affects electoral outcomes only through its interaction with the probability of victory, while neither party enjoys a direct strategic advantage.

Asymmetric Clientelism $z_i^L \neq z_i^R$: When parties exert different levels of clientelistic pressure, equation 13 applies directly. Electoral outcomes are then shaped by both voter loyalties and asymmetric in clientelistic mobilisation.

3.4.1 Thresholds for the choice function Θ_i

The model is built on the endowment effect in political choice. Since elites possess a higher initial endowment ($\omega_e > \omega_c$), their marginal utility is lower compared to non-elites. As a result, elites' welfare is less sensitive to the redistribution transfers or material policy platform (q_e^p) offered by the political parties. Elites derive relatively less utility from marginal increases in income, and therefore do not condition their voting behaviour as strongly on party-specific material policies. Instead, they are able to make voting decisions based on non-material considerations, such as ideology, long-run institutional preferences, or maintaining political stability without incurring substantial welfare losses. Hence, a political party's clientelistic control is less effective over the elite voting group.

By contrast, non-elite, who have lower initial endowments, exhibit a higher marginal utility of income, making their support directly contingent on redistributive promises and transfers. Thus, while elites can afford independence in their voting calculus, non-elites remain comparatively more dependent; thus, non-elites are more likely to be influenced by clientelistic pressure.

[5]

Lemma. *When the voters have an endogenous choice function, the wealthy group chooses to be non-instrumental (vote based on their expected payoff) and the poor group chooses to be instrumental (vote based on expected payoff combined with clientelistic pressure).*

Proof :

A voters behaviour when the voter has a choice of choosing type is given by:

$$\Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R] + \varepsilon_i > 0$$

A voter from group i chooses to be instrumental if $(I_i^L - I_i^R > N_i^L - N_i^R)$ and he chooses to be non-instrumental if $(N_i^L - N_i^R > I_i^L - I_i^R)$. The choice Θ_i is given by:

$$\max_{0 \leq \Theta_i \leq 1} \left\{ \Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R] + \varepsilon_i \right\}$$

The optimisation problem reveals a corner (bang-bang) solution, as the objective function is linear in Θ_i. The voters choose either $\Theta_i = 1$ or $\Theta_i = 0$. □

Optimal Θ_i

If the difference between the net benefits from Non-Instrumental support $(N_i^L - N_i^R)$ exceeds the difference in the net benefits from Instrumental support $(I_i^L - I_i^R)$, then the objective function is increasing in Θ_i, implying optimal value is attained at $\Theta_i = 1$. Conversely, if the net benefits from Non-Instrumental is less than net Instrumental benefits $((N_i^L - N_i^R) < (I_i^L - I_i^R))$, the objective function decreases with respect to Θ_i, and thus reaches its maximum at $\Theta_i = 0$. This formulation captures how the relative strength of non-instrumental and instrumental voting influences optimal strategic allocation.

$$\Theta_i^* = \begin{cases} 1, & \text{if} \quad N_i^L - N_i^R > I_i^L - I_i^R \\ 0, & \text{if} \quad N_i^L - N_i^R < I_i^L - I_i^R \end{cases}$$

[5]The above proposition is consistent with previous empirical literature on the Development and Clientelism, See Hicken (2011)

Where

$$N_i^L - N_i^R \equiv u(\omega_i + q_i^L) - u(\omega_i + q_i^R)$$

and

$$I_i^L - I_i^R \equiv \gamma_L z_i^L [u(\omega_i + q_i^L) - u(\omega_i)] - (1 - \gamma_L) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)]$$

The value of ω_i is publicly known and accessible to all agents. Consequently, voters make decisions based on their initial endowment, while political parties determine the equilibrium allocation $\{q_i^p\}$, taking into account the information available to voters. The equilibrium quantity of transfer q_i^{p*} is determined by maximising the quasi-utilitarian objective function subject to budget constraints and non-negativity constraints.

Proposition 2.1: *There exists an optimal initial endowment level at which instrumental voters transition to non-instrumental behaviour.*

Since the expected utilities are continuous in initial endowment, the intermediate value theorem implies the existence of a threshold level of initial endowment at which voter behaviour transition from instrumental to non-instrumental. This threshold is defined by the point at which the expected utility from being a non-instrumental voter is exactly the same as the expected utility from being an instrumental voter. Below this threshold voters behave instrumentally, responding to both transfers and clientelistic pressure; above it, they become non-instrumental, which their behaviour determined solely by expected utility from transfers. (Proof in Appendix A.7)

Proposition 2.2 : *Clientelistic control is effective only when the voter base is economically disadvantaged; if the electorate is affluent, the political party lacks the capacity to sustain clientelistic influence.*

The proposition highlights that the effectiveness of clientelistic control depends on the economic position of the electorate. Since non-instrumental voters do not condition their voting decisions on clientelistic pressure, changes in the level of such pressure do not affect the elite's electoral choice. Consequently, parties seeking to secure elite support must instead influence their expected utility through the benefits associated with the policy platform. The proof formalizes this distinction by defining a net clientelistic function and identifying a threshold level of initial endowment below which clientelistic influence is feasible and above which it cannot affect electoral behaviour. (Proof in Appendix A.8)

Corollary 2. *Propositions 2.1 and 2.2 together imply that the poor group ($\omega_i < \omega_i^*$) lies in the region where clientelistic control is both preferred by the voter and feasible for the party, while the wealthy group ($\omega_i > \omega_i^{**}$) lies in the region where clientelism is neither preferred nor feasible.*

Propositions 2.1 and 2.2, together with the preceding corollary, provide a possible explanation for why political parties may benefit from a an economically disadvantaged electorate. When voters are poor, their electoral choices are responsive to clientelistic pressure, allowing parties to influence voting behaviour through targeted conditional transfers. This creates scope for parties to use clientelism as an instruments of electoral control while allocating transfers below the level that would arise in the absence of such political considerations. By contrast, as voters becomes more affluent, their electoral decisions becomes less responsive to clientelistic incentives, reducing the effectiveness of this form of political influence. The results therefore suggest that economic disadvantage can create the conditions under which clientelistic practices become electorally valuable, potentially allowing parties to sustain political support with out fully compensating voters thrught transfers.

Proposition 3 : *In an unequal society, elections exacerbate inequality in equilibrium, as the political welfare weight assigned to the elite exceeds that allocated to the non-elite.*

Proof :

The quasi-utilitarian utility function is given as:

$$\sum_{i \in \{e,c\}} \mu_i \sigma_i [\Theta_i + (1 - \Theta_i)\gamma_p^* z_i^p] u(\omega_i + q_i^p)$$

The political welfare weight is:

$$\delta_i = \sigma_i [\Theta_i + (1 - \Theta_i)\gamma_p^* z_i^p]$$

Keeping swing parameter as small and same across all groups ($\sigma_e = \sigma_c = \sigma$), and substituting the value of $\Theta_e = 1$ and $\Theta_c = 0$ into the equation for δ_i we get:

- $\delta_e = \sigma$
- $\delta_c = \sigma\ \gamma_p^*\ z_c^p$ where $0 < \gamma_p < 1$ and $z_c^p = 1$

$$\therefore \delta_e > \delta_c \implies q_e^* > q_c^*$$

The political parties assign a higher welfare weight to the elite group, as a result the equilibrium goods transfer policy favour elite over non-elite. The final endowment of elite is higher than the final endowment of non-elite $(\omega_e + q_e^*) > (\omega_c + q_c^*)$. The positive change in endowment is not proportional as $(\omega_e > \omega_c)$ and $(q_e^* > q_c^*)$.

The analysis above shows that, in the presence of clientelism at lower level of democracy, electoral competition in unequal society leads to an equilibrium that exacerbates inequality. □

3.4.2 Interpretation of equilibrium

Proposition 1 implies that when both parties assign identical welfare weights to elites and, similarly, identical weights to non-elites, the resulting equilibrium generates indifference across parties within each group. In such setting, electoral outcomes are determined by tie-breaking factor of voter loyalty for symmetric clientelism case or by the factor of loyalty and clientelistic weight in case of asymmetric clientelism. With clientelism, however, parties can expand their vote share through two channels: intensifying clientelistic pressure or increasing electoral transfers. Since non-elites are poor, possess high marginal utility from transfers, and form the majority, augmenting large transfers is unsustainable, making clientelistic pressure the more viable strategy. Elites, by contrast, are wealthy, hold low marginal utility from transfers, and make up a minority. Their votes cannot be secured through clientelistic pressure, but rather through private transfers sufficient to offset their limited utility gain. The optimal allocation therefore maximises budgetary resources devoted to elite transfers while maximising clientelistic leverage and minimising resources promised over the non-elite electorate.

Lemma establishes a threshold for the endogenous choice function for the voter group, followed by two parallel propositions. Proposition 2.1 proves the existence of an optimal initial endowment below which voters are instrumental and above which the voters are non-instrumental. Proposition 2.2 deals with the effectiveness of clientelism and proves that clientelism is only effective if the voting population economically is disadvantaged. Proposition 3, which states the nature of equilibrium after a democratic process in a society with inequality and clientelism. The equilibrium from the above propositions is the following:

In an unequal society, the democratic process of election leads to an equilibrium in which political parties maximise clientelistic control over the citizenry while minimising it over the elites. This dynamic results in greater welfare weights being assigned to elites, resulting in higher transfers directed toward them, and ultimately deepening inequality.

Citizens receive the minimum feasible transfers, as their instrumental voting behaviour is shaped by the endogenous voter type choice and the clientelistic control parameter, which makes them responsive to the material incentives. Whereas the elites obtain the maximum feasible transfers, as securing their support requires non-instrumental means such as ideological alignment or policy capture.

3.5 Elite capture and Clientelism

In the previous section, we saw the case for pure clientelism, where in equilibrium political parties assign a higher welfare weight to the elites, leading to higher transfers to the elite voting group, further increasing the consumption inequality in the society. Incorporating the notion of elite capture serves as a robustness check for the existing model developed previously. Elite capture occurs when the local elites finance the electoral campaign of the political parties in exchange for modifying the proposed allocation made by political parties. This is how the political parties get captured by elites. The process of negotiation is along the lines of Grossman and Helpman (1996). The theory shows that: (1) Utility is quasi-linear; (2) The lobby makes a take-it-or-leave-it offer of a policy proposal to each party and (3) Party's participation constraint are binding at the optimum, implies, vote share will be unaffected by the deal offered by the elite group. Elites are driven by two main factors to contribute. First, they may seek to influence electoral outcomes, with interest groups benefiting if their support enhances the prospects of the party whose position on flexible policies aligns most closely with their own. Second, elites may aim to shape the party platform itself, encouraging candidates to adopt policies that better serve the interests of the elite Bardhan and Mookherjee (2000).

A political party's electoral policy now also involves raising funds from the local elites. The campaign funds are denoted by C^P and are used solely for the purpose of the campaign spending. The political party now has the ability to manipulate the voting behaviour by means of persuasion with the help of campaign finances. The updated voting behaviour when party L spends more than party R is as follows:

$$\Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R] + H(C^L - C^R) + \epsilon_i > 0 \tag{14}$$

Here H is a parameter of effectives of campaign finances in mobilising votes, from Grossman and Helpman (1996). Parameter H will end up determining the extent of local capture. H directly affects the distribution from figure 1. When party L spends more than opposition the

distribution shifts right. This indicates that a higher population will vote for party L, effectively increasing the vote share of party L. Similar effect is observed when party R spends more than party L.

The resulting vote share for party L will be:

$$S_L(\gamma^L; q^L, q^R) \equiv \frac{1}{2} + \sum_i \mu_i l_i \sigma_i + h(C^L - C^R) + \sum_i \mu_i \sigma_i \{\Theta_i \left[u(\omega_i + q_i^L) - u(\omega_i + q_i^R)\right] + (1 - \Theta_i)(\gamma^L z_i^L \left[u(\omega_i + q_i^L) - u(\omega_i)\right] - (1 - \gamma^L) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)]\} \quad (15)$$

Where $h \equiv H \sum_i \mu_i \sigma_i$. since μ_i and σ_i are fixed parameters, therefore H and h are directly proportional Bardhan and Mookherjee (2023). The finite upper bound of the vote share determined in equation 9 still holds with the updated vote share function. The equilibrium vote share is given by:

$$S_L(\gamma_L^*; q^L, q^R) \equiv \frac{1}{2} + \sum_i \mu_i l_i \sigma_i + h(C^L - C^R) + \sum_i \mu_i \sigma_i \{\Theta_i \left[u(\omega_i + q_i^L) - u(\omega_i + q_i^R)\right] + (1 - \Theta_i)(\gamma_L^* z_i^L \left[u(\omega_i + q_i^L) - u(\omega_i)\right] - (1 - \gamma_L^*) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)]\} \quad (16)$$

Party p will win with the same probability γ_p.

Following along the lines of Bardhan and Mookherjee (2023), when both clientelism and elite capture are present, the equilibrium allocation of party p maximises the quasi-utilitarian welfare function.

$$\sum_{i=e,c} \mu_i \sigma_i \left[\{\Theta_i + (1 - \Theta_i)\gamma_p^* z_i^p\} u(\omega_i + q_i^p)\right] + \mu_e \; h \; \gamma_p^* u(\omega_e + q_e^p) \quad (17)$$

Over the set of feasible allocation, taking γ_p^* as given, the equilibrium probability of party p winning.

The political welfare weight for different groups are as follows:

$$\delta_c = \sigma_c \{\Theta_c + (1 - \Theta_c)\gamma_p^* z_c^p\}$$
$$\delta_e = \sigma_e \{\Theta_i + (1 - \Theta_i)\gamma_p^* z_e^p + h\gamma_p^*\}$$

Keeping the swing parameter as a small value and same across all groups ($\sigma_e = \sigma_c = \sigma$), and substituting the value of $\Theta_e = 1$ and $\Theta_c = 0$, we get the welfare weight when elite capture and clientelism is involved.

$$\delta_c = \sigma_c \gamma_p^* z_c^p$$
$$\delta_e = \sigma_e + h\gamma_p^*$$

Proposition 1, Proposition 2.1, proposition 2.2 and proposition 3 hold true for the updated value of the political welfare weight. Comparing with the pure clientelism case, it is observed that the the welfare weight for the elite in this case is higher, leading to higher transfers.

This demonstrates that in an unequal society, the democratic electoral process produces an equilibrium in which political parties maximise clientelistic control over the non-elite population while minimising it over the elites. Simultaneously, when elites capture the party election manifestos, policies favour their interests. This dynamic assigns even greater welfare weights to elites than in the case of pure clientelism, leading to higher transfers directed towards them and, ultimately, further deepening of inequality.

4 Structural Violence

The preceding section demonstrated that, in equilibrium, when the society is unequal ex-ante, both scenarios, pure clientelism and a combination of clientelism with elite capture, inequality increases following elections. Moreover, the increase in inequality is more pronounced in the case where clientelism is coupled with elite capture compared to the pure clientelism scenario. When welfare weights for instrumental and non-instrumental voters are determined endogenously, the democratic process produces greater inequality than the initial distribution of endowments, thereby systematically intensifying disparities.

Without clientelism or elite capture, even an unequal society reaches equality under the optimal allocation, with non-elites receiving larger transfers ($q_e^* < q_c^*$):

$$u(\omega_e + q_e^*) = u(\omega_c + q_c^*).$$

With clientelism ($\delta_e > \delta_c$), Non-elites are instrumental type and elite are non-instrumental.

Transfers then favour elite ($\hat{q}_e > \hat{q}_c$), the utility comparison is as follows:

$$u(\omega_e + \hat{q}_e) > u(\omega_c + \hat{q}_c).$$

Adding elite capture, with the same voter types, preserves this ordering ($\tilde{q}_e > \tilde{q}_c$):

$$u(\omega_e + \tilde{q}_e) > u(\omega_c + \tilde{q}_c).$$

Comparing the transfer allocations across the three regimes, we obtain :

- For elites $-\quad \tilde{q}_e > \hat{q}_e > q_e^*$
- For citizen $-\quad \tilde{q}_c < \hat{q}_c < q_c^*$

In the benchmark, redistribution balances out pre-existing endowments. Once clientelism is introduced, political parties find it optimal to allocate more resources towards elites, as non-instrumental elites require higher transfers to compensate for the weak electoral responsiveness. With the addition of elite capture, elite further influence party platforms in their favour, intensifying this skew. Consequently, elites consistently receive more, while citizens systematically loose out.

The citizen's loss in utility related to the egalitarian benchmark as the gap between the utility under the egalitarian allocation and the utility under alternative regimes.

$$\Delta u_c^{Clientelism} = u(\omega_c + q_c^*) - u(\omega_c + \hat{q}_c)$$
$$\Delta u_c^{Clientelism+Capture} = u(\omega_c + q_c^*) - u(\omega_c + \tilde{q}_c)$$

Under the model's equilibrium comparative statics, both gaps are non-negative :

$$\Delta u_c^{Clientelism} > 0$$
$$\Delta u_c^{Clientelism+Capture} > 0$$
$$\Delta u_c^{Clientelism+Capture} > \Delta u_c^{Clientelism}$$

These welfare shortfalls can be defined as structural violence. These systematic welfare losses capture the structural violence embedded deep in democratic electoral dynamics, where struc-

turally conditioned deprivation prevents citizens from realising their potential utility.

The literature in sociology and anthropology provides a term for this form of systemic harm. Galtung (1969) defines 'structural violence' as, " the unintended harm done to human beings through processes that work slowly to erode human life". Similarly, Gupta (2012) characterises this form of violence as structural precisely because it cannot be attributed to any single identifiable actor; rather, it is impersonal, embedded within, and perpetuated by institutional structures themselves.

The analysis developed in this paper draws a close parallel to Gupta's conception of structural violence, as the systemic welfare shortfalls identified here align with his notion of institutionally embedded harm that emerges not from individual actors but from the very structure of governance.

In context of the model presented above, structural violence is the welfare shortfall experienced by the non-elite, i.e., the gap between the potential utility they would attain under a perfectly functioning egalitarian democracy and the actual utility they attain when the democratic process is distorted by clientelism and elite capture.

5 Structural Estimation

The empirical paper by Bardhan et al. (2024) show credible evidence of presence of clientelism rather than programmatic politics. To show the presence of clientelism. The authors use I.V. estimation technique by making use of shock to electoral competition induced by exogenous redistricting of villages which resulted in upper tier governments manipulating allocation across local governemnt only for excludable benefit programs. However, this approach approach identifies the presence of clientelistic behaviour without directly recovering the underlying parameters governing the intensity of clientelistic pressure or the distributional preferences of political parties.

By using the same data, I complement this reduced form evidence with a structural estimation of the model developed above. The structural approach allows me to quantify clientelistic pressure across different levels of electoral uncertainty and to recover the underlying political welfare weights that determine how parties value the welfare of different groups once in office. The objective is therefore not to establish a causal effect of clientelism, but to recover the unobserved structural parameters governing voter behaviour and political parties' allocation

decisions. These parameters characterise the equilibrium mechanism of the model and allow the analysis to extend beyond the observed variation in the data to evaluate counterfactual and out of sample equilibria.

I estimate the model under two alternative specifications. The first treats the parameters governing clientelistic pressure and voter choice as exogenous, closely following the formulation in Bardhan and Mookherjee (2023). This specification provides a benchmark for the case where clientelistic pressure and voter choice is endogenized in accordance with the model developed above (Section 3). In this specification, political parties choose the intensity of clientelistic pressure as a function of voters' initial endowment, while voters electoral responses are themselves determined by their economic circumstances. This allows the estimation to recover the interaction between voters' economic position, clientelistic pressure and political parties' distributional preferences.

The two specifications therefore differ in the extent to which the key behavioural parameters are allows to respond endogenously to economic conditions. Comparing the resulting estimates provide a way to assess how the structural interpretation changes when clientelistic pressure and voter behaviour are explicitly linked to initial endowments. To facilitate comparison across specifications, the estimated utility terms are expressed in log normalised form.

6 Context and Data

This analysis relies on secondary data published in Bardhan et al. (2024), The survey obtains information on benefits received by households and a proxy measure of voting behaviour from a household survey carried in 2011. It includes 2383 households from 88 villages in 57 GPs spread through all districts of West Bengal, except Calcatta and Darjeeling (owing to the relative lack of agricultural occupations in these two districts). The same households were surveyed in an earlier round in 2004. The analysis depends on 2011 survey rounds. The survey characterised households based on indicators.

6.1 Administrative Hierarchy

India is a federal state with legislative, administrative and executive power divided between the centre and state governments. The 73rd amendment to the Constitution established a constitutional framework for the devolution of powers and responsibilities to local governments at the grassroot level. It provided at a three tier system of Panchayati Raj Institution, compris-

ing the Zilla Parishad at the district level, The Panchayat Saminit and Block level, The gram Panchayat, as the lowest tier of this system, is responsible for implementing local development programmes and allocating devolved resources within its jurisdiction. The devolution of functions and financial responsibilities gave Gram Panchayats discretion over the implementation and allocation of certain public/private resources within their jurisdiction. Figure 3 shows how funds are distributed across different level.

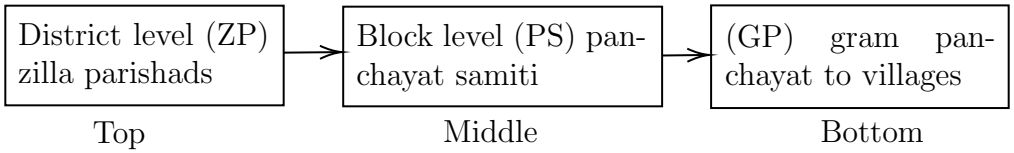


Figure 3: Allocation of Funds

The empirical paper by Bardhan et al. (2024) from where the data is taken focuses on the lower two tiers of local governance, the Panchayat Samiti (PS) and Gram Panchayat(GP).

For the structural estimation exercise, I focus on the lowest tier of local governance hierarchy, the Gram Panchayat (GP). There are three reasons for focusing on this level. First, as discussed, funds are devolved through the different tiers of local government before reaching gram panchayats. They have greater discretion over the allocation these resources to households. This provides a natural setting for examining how the allocation of private benefits can generate a clientelistic channel. Second, the Gram Panchayat constitutes the smallest administrative unit of local governance. The relatively close relationship between elected representative and households within a GP makes it plausible that representative possess information about household characteristics, allowing the allocation of benefits to be targeted according to household level economic characteristics, as assumed in the theoretical framework. Third, West Bengal is among the few Indian States where GP elections are formally contested on party labels Chau et al. (2021). This feature provides a clear link between electoral competition and party affiliation at the lowest level of government, making the GP an appropriate institutional setting for analysing how political parties use electoral strategies to influence local policy and resource allocation.

Political environment: During the period of study (2004 - 2011), there were two principal political alliance broadly based on ideology in West Bengal, the Left Front (LF), a coalition lead by the Communist Party of India (Marxist) and the Non-Left Front, a coalition lead by the All India Trinamool Congress (TMC). The Left Front dominated elected offices corresponding to Gram Panchayat, Panchayat Samiti, District and the state government from 1977 to 2011

and lost its majority in the state assembly to the TMC in 2011.

The 2008 local body elections represented a pivotal moments in West Bengal's electoral politics, as it marked the weakest performance of the left front in three decades. Overall, at district Level, the LF had won all but two districts (Malda and Murshidabad) in 2003, in the next election the LF lost four more districts (Malda, North Dinajpur, Congress, East Medinipur and South 24 Parganas) to the TMC and Congress. The LF majority moved down to 69.25% from 86.82%.

At Panchayat Samiti level, the LF won 280 out of 329 (85%) in 2003, this tally reduced to 190 out 329 (57.45%) in 2008. And at Gram Panchayat level, the lowest level of panchayat hierarchy, Of the 3,220 gram panchayats the LF won 71.77% in 2003 which reduced to 50.47% in 2008 Ghosh (2008); Bardhan et al. (2009). The survey records the political party in each

		Official Election Results		
Tier	**Indicator**	**2003**	**2008**	**change (pp)**
Zilla Parishad	Districts Controlled	15/17 (88.2%)	13/17 (76.5%)	−11.8
Panchayat Samiti	PS Controlled	280/329 (85.1%)	190/329 (57.8%)	−27.4
Gram Panchayat	GPs Controlled	2311/3220 (71.8%)	1625/3220 (50.5%)	−21.3

Table 1: Erosion of Left Front across the three panchayat tiers, West Bengal 2003 vs 2008.

GP in 2003 and 2008. For the empirical analysis, I classify households according to whether they belong to GP governed by Left Front Coalition or by a non-Left coalition.

Given the institutional and electoral politics setting, estimation of clientelistic parameter and political welfare weight needs to be estimated region wise, where a region is denoted by which political alliance came into power at gram panchayat level post 2008 local body elections.

6.2 Determining the initial endowment

The theoretical framework classifies households according to their endowment of wealth and physical capital. In the data, households characteristics include cultivator status, landownership, SC/ST status, and education. Cultivator and landowner status are used to capture differences in physical wealth and capital, while SC/ST status and education capture dimensions of social and human capital, respectively. Since the model focuses on the allocation of

transfers according to initial economic endowments, cultivator and landowner households are classified as having a higher initial endowment and are therefore designated as elites. Households that are neither cultivator nor landowners are classified as non-elites. This classification allows the empirical analysis to map the households characteristics observed in the data onto the two group structure of the theoretical model. When the entire sample in divided into

		Private benefit 04-06		**Private benefit 07-08**		**private benefit 09-11**	
Indicator	Freq	0	≥ 1	0	≥ 1	0	≥ 1
Cultivator and landowner	450	347 (77.11)	103 (22.88)	352 (78.22)	98 (21.77)	354 (78.66)	96 (21.33)
The Rest	1952	1516 (77.66)	436 (22.33)	1603 (82.12)	349 (17.87)	1603 (82.12)	349 (17.87)

Table 2: Private benefits to cultivator and landowners.

'cultivator and landowner households' and 'the rest' and is compared to the amount of private benefits. We see that higher percentage of cultivator and landowner households receive benefits consistently across different time periods.

A natural question is why cast is not incorporated into the distinction between elites and non-elites. This reflects the specific historical context of West Bengal. Following the implementation of extensive land redistribution and agrarian reform under the Left Front, which governed the state for more than three decades, the relationship between caste and landownership was weakened relative to other states. Consequently, we use landownership and cultivator status to identify the economic elite in the empirical analysis, rather than incorporating caste directly into the elite classification. This does not imply that caste and economic status are generally unrelated in India. In several other states, caste remains strongly correlated with landownership and local economic power. Anderson et al. (2015) show that Maratha cast identity in Maharashtra is closely associated with landownership and that clientelistic benefits to non-elites can operate through these locally dominant landed groups. Thus, while caste is not explicitly incorporated into our definition of elites in the West Bengal setting, the broader mechanism through which economic elites mediated clientelistic relationships with non-elites remains relevant.

The household survey asked the heads to report the major benefits they received from the local government. Specifically, they were asked if their household had received benefits from different programs in each year between 2004-2011. Many of the programs listed created benefits that

were clearly private (i.e., household specific) in nature, (a) including employment in programs such as Sampoorna Grameen Rozgar Yojana, MGNREGA, MPLADS; (b) minikits providing farmers with seeds and fertilizers at highly subsidized rates; (c) subsidized credit; (d) land reforms consisting of receipt of land titles or registrations of tenancy contracts; (e) house, toilet - A lump sum transfer to households for house or toilet construction and; (f) Below Poverty Line cards, which identify poor households and entitle them to subsidized for grains and other household items.

For the purpose of structural estimation, I use data on the allocation of private benefits to households following the 2008 local body elections. While clientelistic exchanges are not directly observed in the data, the allocation of these private benefits provides empirical counterpart to the clientelistic resource allocation described in the theoretical framework. This allows the underlying structural parameters governing political welfare weights and clientelistic pressure to be recovered from observed outcomes.

Based on the electoral and household characterisation, it is appropriate to partition the sample along two key dimensions: political alignment and household type. This yields four sub samples: Non-Elite Non-Left, Elite Non-Left, Non-Elite Left, Elite Left. This classification allows for a more precise examination of how clientelistic strategies and resource allocation vary across both political and socioeconomic categories.

6.3 Descriptive statistics

The following section presents relevant patterns from the data Table 3, presents descriptive

group						
Elite	region	N(hh)	$\bar{\sigma}_i$	μ	private benefits	$mean(\omega + q)$
Non-Elite	Non-Left	861	0.3519	0.361	15.79%	1.21
Elite	Non-Left	186	0.3675	0.078	18.21%	2.26
Non-Elite	Left	1073	0.5869	0.450	19.66%	1.25
Elite	Left	262	0.5661	0.110	23.66%	2.28
N = 2383						
$\bar{\sigma}$ is the swing propensity imputed from data (Details in Appendix).						

Table 3: Descriptive stats

information on groups including the number of households, their swing propensity,

The descriptive statistics presented in table indicate that on average, elite households receive a greater volume of private benefits than non-elite households. This observed disparity in transfer

allocations provide the empirical basis for the structural estimation exercise. By combining these observed outcomes with the equilibrium conditions of the model, this analysis recovers the underlying deep parameters governing clientelistic pressure and political welfare weights.

	Average Per-Household Benefits			
	Non-Left GP		Left GP	
	Non-Elite	Elite	Non-Elite	Elite
Private Benefits	0.2137	0.2634	0.2507	0.2824
Employment*	0.0708	0.0914	0.1025	0.1259
Mini kits	0.0151	0.0914	0.0261	0.0649
Housing	0.0256	0.0161	0.0224	0.0305
BLP cards	0.0639	0.0269	0.0801	0.0343
Drinking water	0.0325	0.0376	0.0158	0.0229
Credit	0.0058	0	0.0037	0.0038

**Employment* Includes panchayat provided employment, MGREGA, and MPLAD employment

Table 4: Summary statistics : GP disbursed benefits

Table 4 provides average level of different benefit programs by groups. We observe a consistent pattern with the results where elite household on average receive more private benefits than the non-elite households

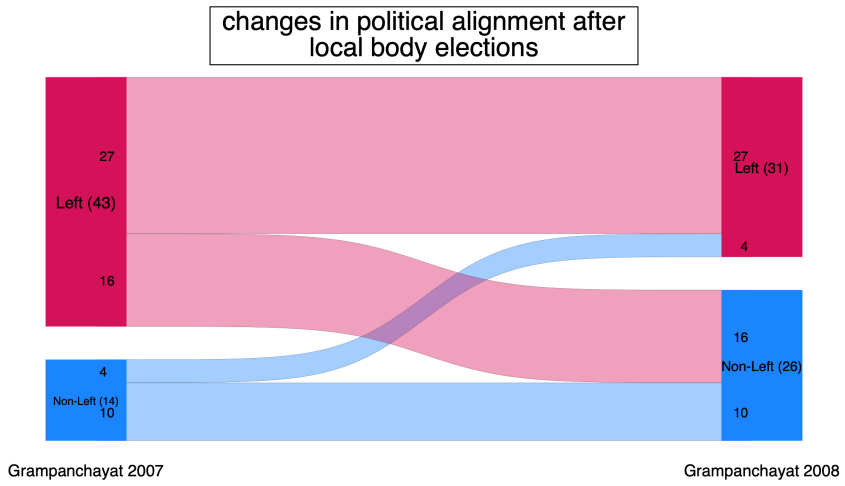


Figure 4: GP switching in the 2008 local body election

Figure 4 shows information from the household survey data, it shows reduction in GPs controlled by LF between the two elections. About 16 GPs moved away from LF majority to the Non-Left majority. At the same time Left Front maintained the overall majority reflecting the overall poll results from the state.

7 Theoretical model and structural reduction

The following section translates the theoretical model form section (REF) into a structural model that incorporates household characteristics and the political affiliation of the GP where the households is situated.

7.1 Model setting

Consider a village economy in which each household i is characterized by its initial endowment of physical capital, $g \in \{E, NE\}$, and the region to which it belongs, $r \in \{L, NL\}$. Households are therefore classified into one of four groups, $G(g,r) \in \{(NE,NL),(E,NL),(NE,L),(E,L)\}$, where E and NE denote elite and non-Elite households, respectively, and L and NL denote Left and Non-Left regions. Elite households are characterized by a higher ex-ante endowment of physical capital, whereas non-elite households have a lower ex-ante endowment.

The ex-ante endowment of household i is denoted by ω_i and is observable to all households and political parties. Let $\mu_{G(g,r)}$ denote the fraction of households belonging to group $G(g,r)$. Elites constitute a small share of the population than non-elites in both regions, such that $\mu_{E,r} < \mu_{NE,r}, \quad r \in \{L, NL\}$.

Following the election, the local government allocates private goods to households as part of the clientelistic exchange underlying electoral support. Let q_i^P denote the transfers received by household i when political party $p \in L, NL$ wins the election. Transfers are subject to a region specific government budget constraint, with separate budget constraints defined for the Left and Non-Left Region.

$$\begin{gathered} B_{NL} + B_L = B, \\ \sum_{i \in \{NE,E\}} \mu_{i,NL}\, q_{i,NL} \leq B_{NL} \qquad ; \qquad \sum_{i \in \{NE,E\}} \mu_{i,L}\, q_{i,L} \leq B_L, \\ \sum_{r \in \{NL,L\}} \sum_{g \in \{NE,E\}} \mu_{r,g}\, q_{r,g} \leq B. \end{gathered} \tag{18}$$

Household in group-region i ends up with utility $u(\omega_i + q_i)$, where u is strictly increasing, differentiable and strictly concave. For the purpose of structural estimation I would be using utility preference to be logarithmic.

$$u(\cdot) \equiv ln(.)$$

7.2 Quasi-Utilitarian Allocation

The government chooses transfers $q_{g,r}$ to maximise the quasi-utilitarian objective

$$W = \sum_{r\in\{L,NL\}} \sum_{g\in\{E,NE\}} \mu_{g,r}\, \delta_{g,r}\, u(\omega_{g,r} + q_{g,r}),$$

subject to a budget constraint in each region (Appendix B.1.1). Both groups in a region share the same budget multiplier, so it cancels from the ratio of their first-order conditions. With log utility, the within-region ratio of welfare weights is

$$\frac{\delta_{E,r}}{\delta_{NE,r}} = \frac{u'(\omega_{NE,r} + q_{NE,r})}{u'(\omega_{E,r} + q_{E,r})} = \frac{\omega_{E,r} + q_{E,r}}{\omega_{NE,r} + q_{NE,r}}, \qquad r \in \{L, NL\}. \tag{19}$$

This ratio depends only on observed incomes and transfers. It is therefore point-identified without any exogenous parameter. The estimates are in Table 5.

Group	δ ratio
$(\frac{elite}{nonelite})_{Left}$	1.8249 (0.0371)
$(\frac{elite}{nonelite})_{Non-Left}$	1.8649 (0.0452)
These ratios hold REGARDLESS of θ or γ.	
They are most robust results of the structural model	
std. errors in parenthesis	

Table 5: Point Identified Political welfare weight ratio's

7.3 Recovering Clientelistic Pressure

In equilibrium, the welfare weight of household i in group (g, r) is

$$\delta_i = \bar{\sigma}_{g,r}[\theta + (1-\theta)\gamma z_i], \tag{20}$$

where $\gamma \in (0.5, 1)$ is the probability that the party wins, z_i is the clientelistic pressure, and θ_i is the weight on non-instrumental voting. Combining (20) with the first-order conditions gives the estimating equation (Appendix B.1.2):

$$\tilde{y}_i = \ln\Big(\theta + (1-\theta)\gamma \exp(x_i'b)\Big) + \varepsilon_i, \qquad z_i = \exp(x_i'b), \tag{21}$$

where $\tilde{y}_i = \ln\lambda_r + \ln(\omega_i + q_i) - \ln\bar{\sigma}_{g,r}$. We estimate b by `GMM`, using the group indicators as instruments (Appendix B.1.3).

We estimate two specifications of the model, which differ in Θ_i and in how z_i varies by group:

- **First Specification** (Bardhan and Mookherjee, 2023): $\Theta_i = \theta$ is exogenous and common to all voters, and $x_i = (1, \text{elite}_i, \text{leftgp}_i, \text{interact}_i)'$. $z_{g,r}$ is the clientelistic parameter experienced by the voting population
- **Second Specification**: Θ_i is chosen by the voter. Poor households vote instrumentally ($\Theta_i = 0$) and rich households do not ($\Theta_i = 1$). Rich households are therefore unaffected by clientelistic pressure, and (21) carries no information about z_i for them. We estimate the extended model on the poor households only for them (21) reduced to $\tilde{y}_i = \ln(\gamma\, exp(x_i'b)) + \varepsilon_i$. Here $x_i = (1,\ \text{Non-Elite}_i \times \text{Left GP}_i)'$. $z_{g,r}$ is the clientelistic pressure set endogenously by the political party after observing the ex-ante endowment of the voting population.

The recovered clientelistic pressures are reported in Tables 6 and 7. For cross model comparison, z is mapped onto $(0, 1)$ with a logistic transformation, consistent with its interpretation as a bounded measure of influence.

Group	Clientelistic Pressure	
$z_{NE,NL}$	$e^{b_0}/(1+e^{b_0})$	$\equiv logit(b_0)$
$z_{E,NL}$	$e^{b_0+b_1}/(1+e^{b_0+b_1})$	$\equiv logit(b_0+b_1)$
$z_{NE,L}$	$e^{b_0+b_2}/(1+e^{b_0+b_2})$	$\equiv logit(b_0+b_2)$
$z_{E,L}$	$e^{b_0+b_1+b_2+b_3}/(1+e^{b_0+b_1+b_2+b_3})$	$\equiv logit(b_0+b_1+b_2+b_3)$
z values are recovered assuming some form of distribution over θ and γ		

Table 6: Recovering clientelistic pressure

Group	Clientelistic Pressure	
$z_{NE,NL}$	$e^{b_0}/(1+e^{b_0})$	$\equiv logit(b_0)$
$z_{NE,L}$	$e^{b_0+b_1}/(1+e^{b_0+b_1})$	$\equiv logit(b_0+b_1)$
z values are recovered assuming some form of distribution over Θ and γ		

Table 7: Recovering endogenised clientelistic parameter

7.3.1 Structural welfare weights.

Substituting the recovered z_i into (20) gives the welfare weight of each group. Because λ_r is common within a region, the within-region ratios $\delta_{E,r}/\delta_{NE,r}$ are identified without knowing the regional marginal value of public funds. This makes them directly comparable to the point-identified ratios in (19).

8 Estimation procedure

The clientelistic pressure parameter b are estimated from equation (21) by the method of `GMM`. The identifying assumption is that a household's group type (elite status and region) is fixed before any measurement or implementation noise is realised. The noise is therefore unrelated to group membership, and the group indicators serve as instruments. In both specifications the number of instruments equal the number of parameters, so the models are exactly identified. Standard errors are clustered at village level. Full details of the estimator, the standard errors and sensitivity analysis are in Appendix: B.1.

8.1 Equilibrium Win Probability

We solve the model separately for each region $r \in \{NL, L\}$, the gram panchayats governed by non-left and Left coalition. Each group $g \in \{E, NE\}$ is described by four observed quantities: its population share $\mu_{g,r}$, mean swing propensity $\bar{\sigma}_{g,r}$, mean loyalty $l_{g,r}$ (set exogenously), and the mean per capital utility gain from its transfers $\Delta u_{g,r} = u(\omega_{g,r} + q_{g,r}) - u(\omega_{g,r})$. The clientelistic pressure for $z_{g,r}(\gamma)$ are the `GMM` estimates from Section (B.1.3), computed for each value of γ.

Following Bardhan and Mookherjee (2023), the win probability γ is a self-fulfilling belief. The probability voters assign to the party winning must equal the probability that the vote share this belief induces actually delivers victory.

We observe only the transfers delivers by the winning party, so we set the losing party's transfers to zero. For clientelistic benefits, which require holding office, this follows from the timing of the model. For programmatic transfers it is an identifying restriction: we measure the gain from the observed transfers relative to receiving nothing, rather than relative to losing party's unobserved offer (Appendix: B.3). Under this restriction the party's vote share in region r is

$$S_r(\gamma) = \frac{1}{2} + A_r + B_r(\gamma), \tag{22}$$

where

$$\begin{aligned} A_r &= \sum_{g \in \{NE,E\}} \mu_{g,r} \bar{\sigma}_{g,r} l_{g,r}, \\ B_r(\gamma) &= \sum_{g \in \{NE,E\}} \mu_{g,r} \bar{\sigma}_{g,r} \Delta u_{g,r} \Big[\theta + (1-\theta)\gamma_r z_{g,r}(\gamma)\Big] \end{aligned} \quad ; r \in \{NL, L\} \tag{23}$$

The loyalty term is A_r is constant, and is set exogenously [6]. If $l_{g,r} = 0$ means no net lean thus $A_r = 0$, if $A_r > 0$ means above average prior support foe the party than won the 2008 election. The clientelistic term B_r weights each group's utility by the same expression that defines its welfare weight in (20). The vote share and the moment condition used to recover z are therefore consistent with each other.

The equilibrium win probability solves

$$\gamma_r = \phi(S_r(\gamma_r)), \qquad \phi(s) = \frac{1}{1 + \exp\big(-\kappa(s - \frac{1}{2})\big)} \tag{24}$$

Since $\phi(1/2) = 1/2$, an even vote split corresponds to an even contest. The slope κ governs how sharply a vote share advantage becomes a near certain victory, and is inversely related to aggregate electoral noise. We set $\kappa = 8$ as the baseline

Although γ and θ both enter the (24), they are different kinds of object. The win probability γ is endogenous and must be self consistent. The non-instrumental weight θ is a structural parameter of the electorate and satisfies no such condition. Equation (24) supplied a single equation, which pins down γ but cannot determine γ and θ jointly. We therefore treat θ as exogenous, solve for γ_r^*, and report how $\gamma_r^*(\theta)$ changes as θ varies.

In the second specification, Θ is determined by household type rather than set exogenously. Rich households vote non-instrumental ($\Theta = 1$) and respond only to the programmatic gain, while poor households vote instrumentally ($\Theta = 0$) and respond only through clientelistic control. The clientelism term becomes

$$B_r^{\text{ext}}(\gamma) = \mu_{E,r}\bar{\sigma}_{E,r}\Delta u_{E,r} + \mu_{NE,r}\bar{\sigma}_{NE,r}\,\Delta u_{NE,r}\;\gamma\; z_{NE,r}(\gamma), \tag{25}$$

where $z_{g,r}(\gamma)$ are the extended model estimates, which are available for poor households only. The fixed (24) then contains no free voting parameter, so γ_r^* is determined without a sensitivity analysis over θ.

8.2 Empirical Predictions

The theoretical framework generates four empirical predictions that guide the structural estimation and subsequent sensitivity analyses. First, clientelistic pressure is predicted to decrease

[6](Appendix REF, measures Loyalties from the pre 2008 voting behaviour and centred)

with the probability of electoral victory. As the probability of winning increases, the marginal electoral benefit of exerting additional clientelistic pressure declines, reducing the incentive for political parties to rely on clientelistic strategies.

Second, clientelistic pressure is predicted to increases in θ. As θ increases, a larger share of voters becomes less responsive to policy incentives, reducing the effectiveness of policy based electoral strategies. Political parties therefore have greater incentives to employ clientelistic pressure to influence the remaining instrumental voters.

Third, the political welfare weight assigned to a group is predicted to increase with the probability of electoral victory and decrease along θ the non-instrumental weight. A higher probability of winning increases the value to a political party of securing the support of a given group. In contrast, as the prevalence of non-instrumental voting increases policy based electoral become less effective, reducing the political value of allocating resources to the group through welfare enhancing policies.

Finally, the model predicts that the equilibrium political welfare weight assigned to elites exceeds that assigned to non-elites,

$$\frac{\delta_e}{\delta_c} > 1.$$

This implies that elites receive a higher political welfare weight in equilibrium and, consequently, a larger share of private benefits relative to non-elites. The resulting allocation therefore provides a mechanism through which electoral competition can sustain, and potentially reinforce, existing economic inequality.

9 Results- First Specification

9.1 Baseline estimates

The table 8 reports the convergence test for the structural estimation. To assess convergence, the weight assigned to non-instrumental voters, θ, and the probability of winning, γ, are held fixed at predetermined values. The two step `GMM` estimation is then performed under these parameter values. The results indicate that the structural equation converges, and the moment conditions are mutually consistent.

The recovered clientelistic pressure reported in Table 8 lie within a narrow bank between 0.796 and 0.935, each precisely estimated and significant at the one percent level. The recovered

Two-step GMM ($\theta = 0.515, \quad \gamma = 0.780$)
Cluster = 88 villages; N = 2382 ; Converged = True

β estimates		*Recovered clientelistic parameter*		*Recovered Welfare weight*		*Ratio of welfare weights*	
Parameter	Mean	Parameter	Mean	Parameter	Mean	Parameter	Mean
β_0	1.9742^{***}	$z_{NE,NL}$	0.8781^{***}	$\delta_{NE,NL}$	0.2981^{***}		
	(0.0316)		(0.0034)		(0.0005)	$ratio_{NL}$	1.0711^{***}
β_1	0.6981^{***}	$z_{E,NL}$	0.9354^{***}	$\delta_{E,NL}$	0.3193^{***}		(0.0016)
	(0.0362)		(0.0018)		(0.0002)		
β_2	-0.6108^{***}	$z_{NE,L}$	0.7963^{***}	$\delta_{NE,L}$	0.4791^{***}		
	(0.0597)		(0.0082)		(0.0018)	$ratio_L$	1.0112^{***}
β_3	0.1418^{**}	$z_{E,L}$	0.9005^{***}	$\delta_{E,L}$	0.4844^{***}		(0.0034)
	(0.0553)		(0.0017)		(0.0004)		

Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. $0\ vs\ \beta$; $0.5\ vs\ z$; $0\ vs\ \delta$; $1\ vs\ ratio$

Table 8: Convergence Test (Sanity Check)

clientelistic pressure for elite is strictly higher than non-elites regardless of which political party controls the village. Within each voter type, pressure is higher in non-Left gram panchayats than in left control ($z_{NE,NL} > z_{NE,L}$ *and* $z_{E,NL} > z_{E,L}$).

Translating these pressures into the party's implicit welfare weights, the two within region ratios are the economically interpretable quantities, since they cancel the region specific budget multiplier λ_r. Both ratios are greater than one, implying that policy preference of both parties is regressive in nature.

Relative to a utilitarian benchmark this represents mild regressivity, the elite advantage is approximately seven percent in non-left gram panchayats and on percent in left controlled ones, both statistically significant at one percent level.

It should be emphasised that, because the system is exactly identified, four instruments comprising the constant, elite, left-control, and interaction terms against four structural parameter. Convergence confirms the existence of an interior solution that reproduces the four group region mean at the benchmark values of θ and γ, rather than testing any overidentifying restrictions. The convergence result therefore support the numerical stability of the estimation procedure, though it does not by itself constitute a test of specification validity.

9.2 Sensitivity to γ

Since the `GMM` estimation converges when (θ) and (γ) are held fixed, the next step is to examine the sensitivity of the recovered structural parameters to the assumed probability of winning. Because (γ) is not directly observed, the estimation is repeated over a grid of values, $(\gamma \in (0.5, 1))$. This exercise allows us to assess whether the estimated welfare weights and clientelistic pressure are sensitive to alternative assumptions about the electoral prospects of a political party.

Table **??** reports the results. The sensitivity table shows that clientelistic pressure experienced by the households declines as the electoral victory becomes certain. However the relative difference across groups does not change.

The political welfare weight estimated from the structural equation (20), rises across all four groups. As electoral victory becomes more certain, a given observed allocation is rationalised by a smaller clientelistic pressure but at a larger political welfare weight, since the instrumental value of a monitored transfer falls when its delivery is more assured. The decline in clientelistic parameter with γ is consistent with the theoretical prediction that the incentive to exert clientelistic pressure is strongest with high electoral competition and low with low electoral competition. Throughout the range of γ, elites retain higher recovered pressure and higher welfare weights than non-elites within each region.

The within group welfare weight ratios represent the distributive policy preference of the political parties. The preferences in table **??** show that non-left political party's preference is regressive in nature regardless of probability of winning. Where as villages where Left came into power estimated to have progressive preference at high level of electoral uncertainty 0.989 at $\gamma = 0.51$. As the uncertainty reduces, the estimated preference for the left moves to being regressive $1.009 - 1.029$ when the probability of winning increases from $0.75 - 0.99$.

These patterns are displayed graphically in Figure 5 and Figure 6. The left panel from Figure 5 plots the recovered clientelistic pressures against γ, the four series decline monotonically. Figure 6 plots the two welfare weight ratio, both are nearly flat in γ, with the non-Left ratio lying above the left ratio.

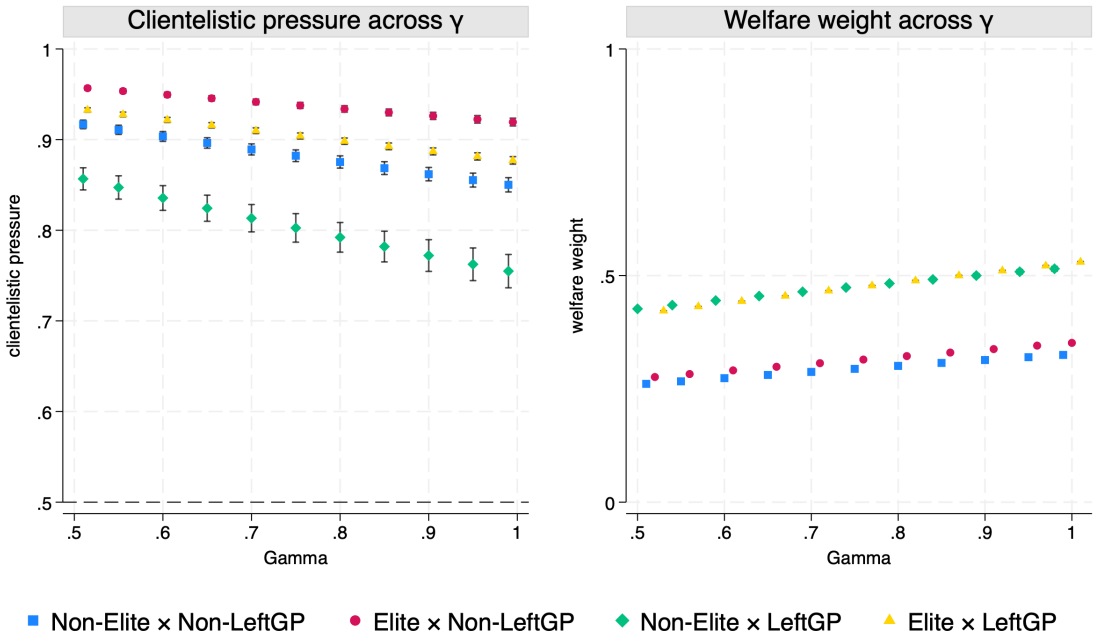


Figure 5: Clientelistic pressure and welfare weight across γ.

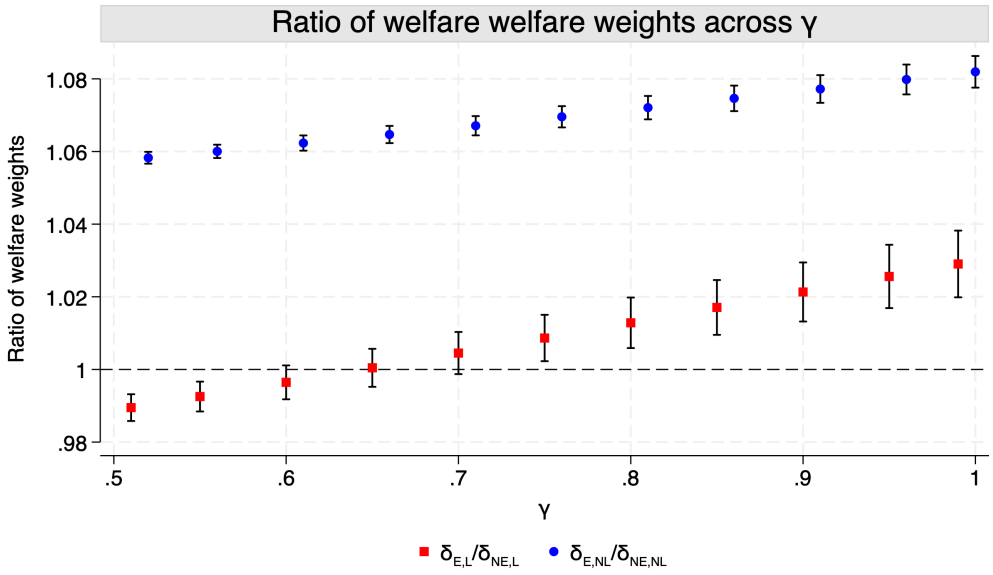


Figure 6: Ratio of welfare weight across γ

9.2.1 Fixed point mapping

The sensitivity analysis leaves γ undetermined within the admissible range. To resolve this, the winning probability is determined endogenously through fixed point iteration from subsection (REF), which selects the value of each γ in each region consistent with the model's equilibrium fixed point condition. Table 9 reports the estimates evaluated at these self-consistent winning probabilities.

Two step GMM ($\theta = 0.515$); Cluster village = 88; Converged = True							
Left Region				Non-Left Region			
recovered *Clientelistic parameter*		*recovered welfare wt.*		recovered *Clientelistic parameter*		recovered *welfare wt.*	
parameter	mean	parameter	mean	parameter	mean	parameter	mean
$\gamma_L^* = 0.8047$				$\gamma_{NL}^* = 0.6906$			
$z_{NE,L}$	0.7912^{***}	$\delta_{NE,L}$	0.4835	$z_{NE,NL}$	0.8905^{***}	$\delta_{NE,NL}$	0.2862
	(0.0084)		(0.0019)		(0.0031)		(0.0004)
$z_{E,L}$	0.8977^{***}	$\delta_{E,L}$	0.4899	$z_{E,NL}$	0.9424^{***}	$\delta_{E,NL}$	0.3053
	(0.0018)		(0.0004)		(0.0016)		(0.0002)
ratio		1.0132^{***}		ratio		1.0666^{***}	
		(0.0036)				(0.0013)	
$***p < 0.01, **p < 0.05, *p < 0.10.$							

Table 9: Result from contraction mapping

The fixed point iteration resolves the indeterminacy in γ by requiring the probability of winning used by voters to evaluate the credibility of clientelistic promises to coincide with the probability generated by the equilibrium. The resulting equilibrium probabilities $\gamma_L^* = 0.8047$ and $\gamma_{NL}^* = 0.6906$. The higher probability in Left controlled GP's indicate lower competition, while the lower probability in Non-Left GPs reflects a high competition electoral environment.

This difference is reflected in the estimated clientelistic pressure. Although clientelistic pressure is higher in Non-Left GPs, where electoral security is lower, the higher pressure can be interpreted as an effort to compensate for weaker institutional control. In contrast, greater electoral security reduces the need for intensive clientelistic conditioning. The welfare weight ratios reinforce this pattern: The elite advantage is 6.7% in Non-Left GPs compared with 1.3% in Left GPs. Thus, elite preference is stronger where electoral competition is higher.

The alluvial diagram in figure 4 provides additional context for this interpretation The political control changed substantially between the 2007 and 2008 local elections, with the Left's share of GPs falling from 43 to 31 while Non-Left control increased from 13 to 26. The estimated equilibrium therefore reflects an environment of considerable political competition rather than stable partisan control. Taken together, the results suggests that clientelism operate in mechanisms through which political parties secure electoral support. This mechanism is more pronounced in Non-Left GPs, where electoral control is weaker and political competition is higher.

9.3 Simulation over $\theta \times \gamma$

The preceding analysis examined the sensitivity of the structural estimates to γ, holding θ fixed at an exogenously specified value. Next, I relax this assumption by allowing θ to vary and evaluate the estimates over a two-dimensional (θ, γ) grid. This exercise provides a systematic assessment of the sensitivity of the recovered structural parameters to alternative values of both behavioural parameters. Finally, fixed point iteration procedure is applied to jointly recover the equilibrium values γ^* for each θ, together with the corresponding clientelistic pressure $z^*(\theta, \gamma^*(\theta))$.

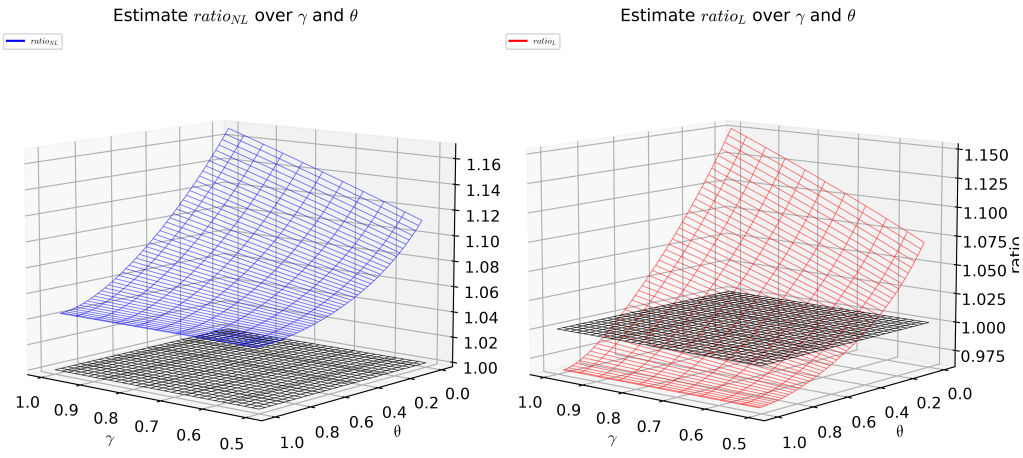


Figure 7: 3-plot of ratio over $\theta \times \gamma$

Figure 7 illustrates this distinction. The Non-Left surface remains above the unit plain throughout the (γ, θ) parameter space, indicating that the estimated elite preference is robust to alternative assumptions about the probability winning and the composition of electorate. The left surface, however, crosses the unit plane. The estimated elite preference is identified only over a restricted set of combinations of γ and θ. This suggests that, conditional on the share of non-instrumental voters and the equilibrium probability of winning, regions governed by the Left-Party could have adopted more egalitarian and progressive policies. Figure 8 provides the corresponding contour representation, showing clearly the combinations of γ and θ where preferences move from progressive to regressive.

9.4 Fixed point Iteration

The equilibrium condition jointly involves γ and θ, but the two parameters play different roles in the model. While γ is endogenous and determined by the electoral equilibrium, θ is a structural parameter capturing the composition of electorate. Consequently, imposing a two-dimensional

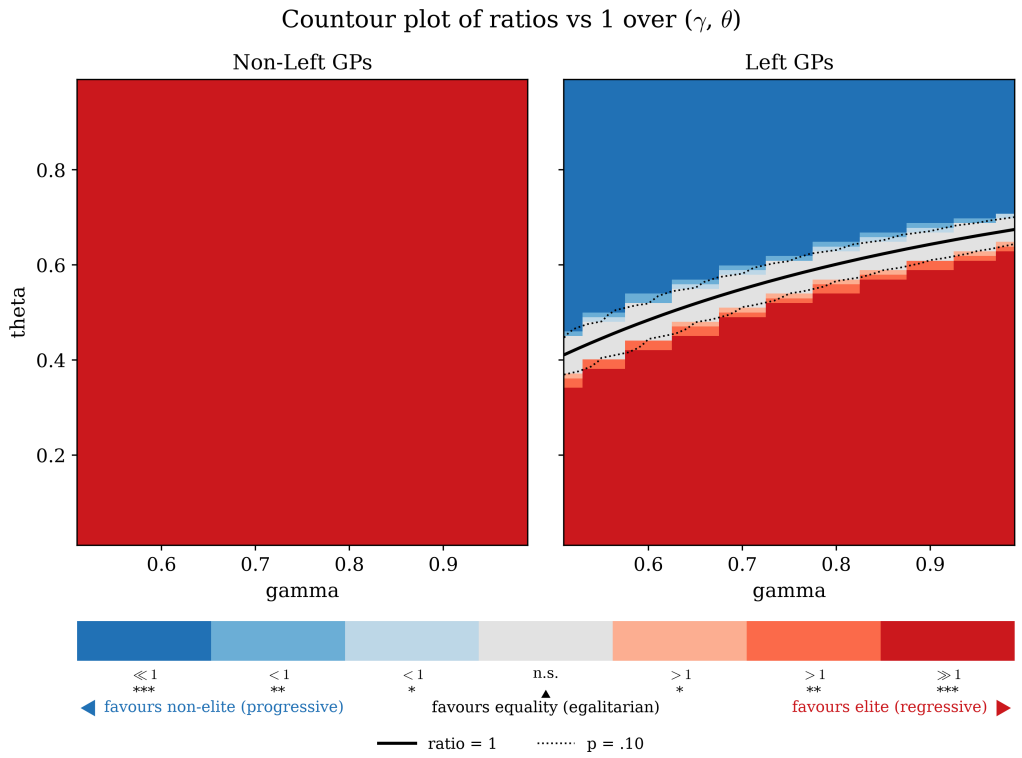


Figure 8: Contour plot of ratio

contraction over (γ, θ) would treat the structural parameter as endogenous and introduce an identifying restriction that is not implied by the model.

I therefore solve the fixed point condition conditional on each value of θ in the parameter grid. for each θ, the contraction mapping yields the corresponding equilibrium, value $(\gamma^*(\theta))$. The resulting equilibrium locus characterizes how the equilibrium probability of winning varies with the weight assigned to non-instrumental voters. Clientelistic pressure is then recovered along this locus as

$$z^*(\theta) = z(\gamma^*(\theta), \theta).$$

This procedure preserves the distinction between the structural composition of the electorate and the endogenous electoral outcome. It also provides a systematic way to examine how changes in the structural parameter θ affect both the electoral equilibrium and the associated level of clientelistic pressure. Political welfare weight and political preference is estimated after equilibrium values of z are recovered.

Table 10 reports these equilibrium values along the resulting $\gamma^*(\theta)$ locus. As θ increases, the equilibrium probability of winning increases in both Non-Left and Left GPs. Clientelistic pressure also increases with θ, consistent with the prediction that a greater weight on non-instrumental voting strengthens the incentives to use clientelistic strategies.

Table 10: Equilibrium values along the locus $\gamma^*(\theta)$, with standard errors.

	Non-Left GPs						Left GPs					
θ	$\gamma^*(\theta)$	z_{NE}	z_E	δ_{NE}	δ_E	δ_E/δ_{NE}	$\gamma^*(\theta)$	z_{NE}	z_E	δ_{NE}	δ_E	δ_E/δ_{NE}
0.010	0.672	0.829***	0.900***	0.198***	0.224***	1.132***	0.776	0.721***	0.837***	0.331***	0.369***	1.116***
		(0.004)	(0.002)	(0.001)	(0.001)	(0.005)		(0.008)	(0.002)	(0.003)	(0.001)	(0.010)
0.100	0.675	0.838***	0.906***	0.214***	0.239***	1.116***	0.782	0.729***	0.845***	0.360***	0.393***	1.093***
		(0.004)	(0.002)	(0.001)	(0.001)	(0.004)		(0.008)	(0.002)	(0.003)	(0.001)	(0.009)
0.200	0.679	0.848***	0.914***	0.233***	0.256***	1.101***	0.788	0.740***	0.856***	0.391***	0.418***	1.070***
		(0.004)	(0.002)	(0.001)	(0.000)	(0.003)		(0.008)	(0.002)	(0.003)	(0.001)	(0.007)
0.300	0.683	0.860***	0.922***	0.250***	0.272***	1.088***	0.793	0.753***	0.868***	0.422***	0.443***	1.050***
		(0.004)	(0.002)	(0.001)	(0.000)	(0.003)		(0.008)	(0.002)	(0.003)	(0.001)	(0.006)
0.400	0.687	0.873***	0.931***	0.267***	0.288***	1.077***	0.799	0.769***	0.881***	0.451***	0.465***	1.032***
		(0.003)	(0.002)	(0.000)	(0.000)	(0.002)		(0.008)	(0.002)	(0.002)	(0.001)	(0.005)
0.500	0.690	0.888***	0.941***	0.284***	0.303***	1.068***	0.804	0.788***	0.895***	0.479***	0.487***	1.016***
		(0.003)	(0.002)	(0.000)	(0.000)	(0.001)		(0.008)	(0.002)	(0.002)	(0.000)	(0.004)
0.600	0.693	0.905***	0.951***	0.299***	0.317***	1.060***	0.809	0.812***	0.912***	0.506***	0.507***	1.001
		(0.003)	(0.001)	(0.000)	(0.000)	(0.001)		(0.008)	(0.002)	(0.002)	(0.000)	(0.003)
0.700	0.697	0.924***	0.962***	0.314***	0.331***	1.054***	0.813	0.841***	0.930***	0.531***	0.525***	0.988***
		(0.002)	(0.001)	(0.000)	(0.000)	(0.001)		(0.008)	(0.001)	(0.001)	(0.000)	(0.002)
0.800	0.699	0.946***	0.974***	0.328***	0.344***	1.049***	0.817	0.880***	0.951***	0.554***	0.541***	0.976***
		(0.002)	(0.001)	(0.000)	(0.000)	(0.000)		(0.007)	(0.001)	(0.001)	(0.000)	(0.001)
0.900	0.702	0.971***	0.986***	0.341***	0.356***	1.046***	0.820	0.930***	0.974***	0.573***	0.555***	0.968***
		(0.001)	(0.000)	(0.000)	(0.000)	(0.000)		(0.004)	(0.001)	(0.000)	(0.000)	(0.000)
0.990	0.704	0.997***	0.999***	0.351***	0.366***	1.044***	0.822	0.992***	0.997***	0.586***	0.565***	0.965***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)		(0.001)	(0.000)	(0.000)	(0.000)	(0.000)

Notes. Each θ block: point estimates on the first line, standard errors in parentheses below. All quantities are evaluated at the equilibrium point (γ^*, θ), where γ^* solves the region-specific contraction mapping (a derived fixed point, no standard error). Stars denote two-sided tests against each quantity's null: z against 0.5, δ_{NE} and δ_E against 0, and δ_E/δ_{NE} against 1; $^*p < .10$, $^{**}p < .05$, $^{***}p < .01$. For the ratio, values above 1 indicate regressive (pro-elite) targeting, below 1 progressive. Slope $k = 8$; γ is the win probability.

The welfare weight ratios reveal an important difference across the two political environments. In Non-Left GPs, the ratio remains above throughout the range of θ, indicating a robust elite advantage along the entire equilibrium locus. In the Left GPs, the ratio is much closer to one and eventually falls below one as θ increases. At the benchmark value $\theta = 0.515$, the equilibrium probabilities are $\gamma^*_{NL} = 0.691$ and $\gamma^*_L = 0.805$, while corresponding welfare weight ratios are 1.067 and 1.013, respectively. Both regions show regressive policy preference.

Figure 9 illustrates this equilibrium locus with the broader (γ, θ) parameter space. The non-left locus lies entirely in the region where the welfare weight ratio exceeds one, whereas the Left locus crosses the unit boundary. This indicates that the elite advantage is a robust equilibrium feature in Non-Left GPs but us sensitive to the equilibrium value of θ and γ in the Left GPs.

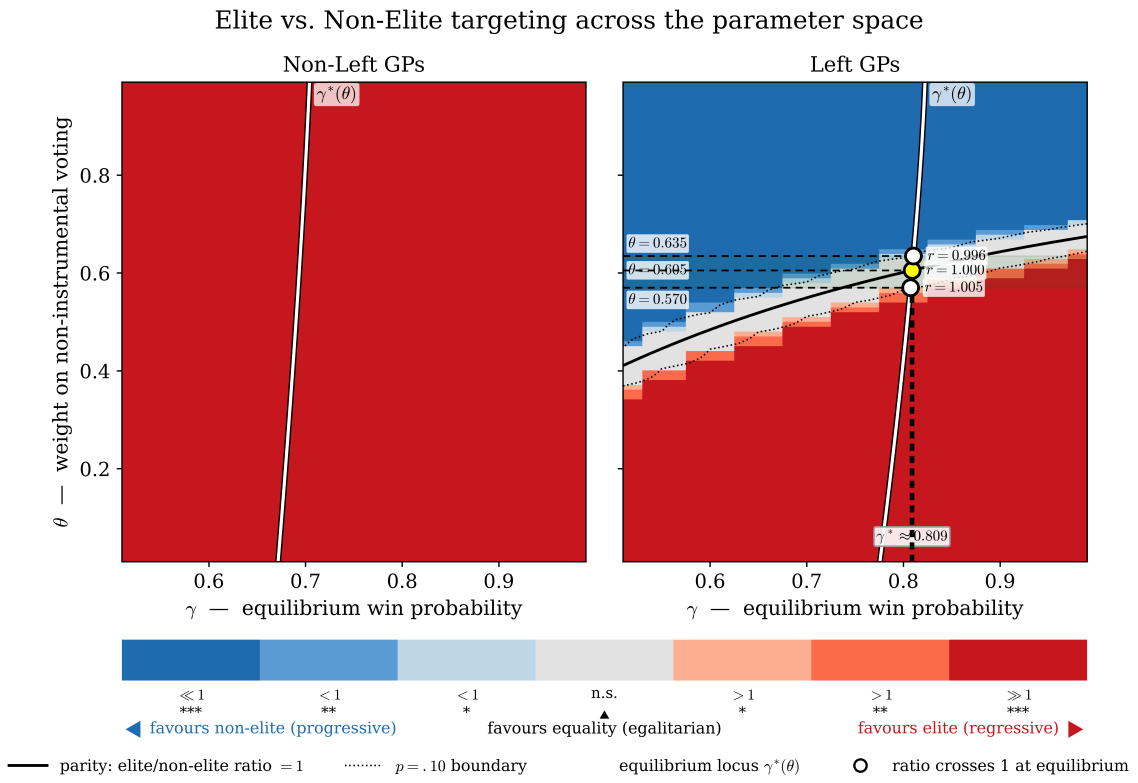


Figure 9: Contraction mapping

10 Results - Second Specifications

Table 11 reports the convergence test for the structural estimation with endogenous voter type Θ and endogenous clientelistic pressure z. The theoretical model predicts that elite voters choose the non instrumental type, implying $\Theta_E = 1$, since their voting decisions is therefore unaffected by clientelistic pressure. Non-elite voters choose the instrumental type, as the expected utility from instrumental voting is strictly greater than from non-instrumental voting, implying that clientelistic pressure has positive effect on voting choice. Political parties consequently have an incentive to exert positive clientelistic pressure on non-elites.

To assess convergence, the probability of winning γ, is held at a predetermined value while Θ is determined endogenously by the model. The two-step `GMM` structural equations converge and that the estimates moment conditions are mutually consistent. The recovered clientelistic pressure on non-elites is 0.8059 in non-Left GPs and 0.7188 is Left GPs, both statistically significant at the one percent level. This clientelistic pressure is higher in Non-Left GPs than in Left GPs.

Two step GMM ($\gamma = 0.78$); Cluster = 88 village level, Converged = True **Θ and z are endogenized**							
b estimates		*recovered clientelistic parameter*		*recovered welfare wt.*		*ratio of welfare wts.*	
parameter	mean	parameter	mean	parameter	mean	parameter	mean
b_0	1.4237	$z_{NE,NL}$	0.8059^{***}	$\delta_{NE,NL}$	0.2212		
	(0.0266)		(0.0042)		(0.0011)	$ratio_{NL}$	1.6613
b_1	−0.4852	$z_{E,NL}$	0.0000	$\delta_{E,NL}$	0.3676		(0.0086)
	(0.0460)		(0.0000)		(0.0257)		
		$z_{NE,L}$	0.7188^{***}	$\delta_{NE,L}$	0.3290		
			(0.0076)		(0.0035)	$ratio_L$	1.7204
		$z_{E,L}$	0.0000	$\delta_{E,L}$	0.5662		(0.0182)
			(0.0000)		(0.0226)		
$***p < 0.01, **p < 0.05, *p < 0.10.$							

Table 11: Convergence Test

Translating the estimated clientelistic pressure into political welfare weights using

$$\delta_{gr} = \bar{\sigma}_{gr}[\Theta_{gr} + (1 - \Theta_{gr})\gamma z_{gr}], \tag{26}$$

the within region welfare weight ratios produce the economically relevant comparison, as the region specific budget multiplier cancels out. Both ratios exceed unity, indicating that elites receive a higher political welfare weight than non-elites in both regions. Moreover, the estimated ratios are close to the point identified ratios reported in Table 5, providing further support for the robustness of the structural estimation.

10.1 Sensitivity to γ

Since the GMM estimation converges when γ is held fixed, the next step is to examine the sensitivity of the recovered structural parameters to the assumed probability of winning. Because γ is not directly observed, the estimation is repeated over a grid of values, $\gamma \in (0.5, 1)$. This allows us to assess whether the estimated welfare weights and clientelistic pressure are sensitive to alternative assumptions about the electoral prospects of a political party.

Table 12 reports the results. In this specifications, only two slope parameters, b_0 and b_1, are estimated. These parameters govern the level of clientelistic pressure on non-elites and its interaction with political alignment. The recovered clientelistic pressure on Non-Elites declines as γ increases in both regions while the political welfare weight assigned to non-elites increases.

Table 12: Sensitivity to γ

	$\beta's$		$z's$				$\delta's$				Ratio's	
γ	β_0	β_1	$z_{NE,NA}$	$z_{E,NA}$	$z_{NE,A}$	$z_{E,A}$	$\delta_{NE,NL}$	$\delta_{E,NL}$	$\delta_{NE,L}$	$\delta_{E,L}$	$ratio_{NL}$	$ratio_L$
0.51	1.849***	-0.485***	0.864***	0.000	0.796***	0.000	0.155***	0.368***	0.238***	0.566***	2.371***	2.375***
	(0.027)	(0.046)	(0.003)	(0.000)	(0.006)	(0.000)	(0.001)	(0.026)	(0.002)	(0.023)	(0.166)	(0.097)
0.55	1.773***	-0.485***	0.855***	0.000	0.784***	0.000	0.165***	0.368***	0.253***	0.566***	2.222***	2.238***
	(0.027)	(0.046)	(0.003)	(0.000)	(0.006)	(0.000)	(0.001)	(0.026)	(0.002)	(0.023)	(0.156)	(0.091)
0.60	1.686***	-0.485***	0.844***	0.000	0.769***	0.000	0.178***	0.368***	0.271***	0.566***	2.063***	2.092***
	(0.027)	(0.046)	(0.004)	(0.000)	(0.007)	(0.000)	(0.001)	(0.026)	(0.002)	(0.023)	(0.145)	(0.086)
0.65	1.606***	-0.485***	0.833***	0.000	0.754***	0.000	0.191***	0.368***	0.288***	0.566***	1.929***	1.968***
	(0.027)	(0.046)	(0.004)	(0.000)	(0.007)	(0.000)	(0.001)	(0.026)	(0.003)	(0.023)	(0.135)	(0.081)
0.70	1.532***	-0.485***	0.822***	0.000	0.740***	0.000	0.203***	0.368***	0.304***	0.566***	1.815***	1.862***
	(0.027)	(0.046)	(0.004)	(0.000)	(0.007)	(0.000)	(0.001)	(0.026)	(0.003)	(0.023)	(0.127)	(0.077)
0.75	1.463***	-0.485***	0.812***	0.000	0.727***	0.000	0.214***	0.368***	0.320***	0.566***	1.715***	1.770***
	(0.027)	(0.046)	(0.004)	(0.000)	(0.007)	(0.000)	(0.001)	(0.026)	(0.003)	(0.023)	(0.120)	(0.073)
0.80	1.398***	-0.485***	0.802***	0.000	0.714***	0.000	0.226***	0.368***	0.335***	0.566***	1.628***	1.690***
	(0.027)	(0.046)	(0.004)	(0.000)	(0.008)	(0.000)	(0.001)	(0.026)	(0.004)	(0.023)	(0.114)	(0.070)
0.85	1.338***	-0.485***	0.792***	0.000	0.701***	0.000	0.237***	0.368***	0.350***	0.566***	1.551***	1.619***
	(0.027)	(0.046)	(0.004)	(0.000)	(0.008)	(0.000)	(0.001)	(0.026)	(0.004)	(0.023)	(0.109)	(0.067)
0.90	1.281***	-0.485***	0.783***	0.000	0.689***	0.000	0.248***	0.368***	0.364***	0.566***	1.483***	1.556***
	(0.027)	(0.046)	(0.005)	(0.000)	(0.008)	(0.000)	(0.001)	(0.026)	(0.004)	(0.023)	(0.104)	(0.065)
0.95	1.227***	-0.485***	0.773***	0.000	0.677***	0.000	0.258***	0.368***	0.378***	0.566***	1.422***	1.499***
	(0.027)	(0.046)	(0.005)	(0.000)	(0.008)	(0.000)	(0.002)	(0.026)	(0.005)	(0.023)	(0.100)	(0.063)
0.99	1.185***	-0.485***	0.766***	0.000	0.668***	0.000	0.267***	0.368***	0.388***	0.566***	1.378***	1.458***
	(0.027)	(0.046)	(0.005)	(0.000)	(0.008)	(0.000)	(0.002)	(0.026)	(0.005)	(0.023)	(0.097)	(0.061)

Standard errors in parentheses.
$***p < 0.01, **p < 0.05, *p < 0.10$.

The welfare weight assigned to elites, however, remains constant across γ. This follows directly from equation (26). Since $\Theta_E = 1$, the welfare weight of elites reduces to their mean swing propensity $\bar{\sigma}_{gr}$, and is therefore independent of γ and z. In contrast, since $\Theta_{NE} = 0$, the welfare weight of non-elites is given by $(\bar{\sigma}_{gr}\ \gamma\ z_{gr})$, making it directly dependent on the probability of winning and clientelistic pressure.

The most important pattern is that recovered clientelistic pressure on non-elites declines as γ increases in both regions. This is consistent with the credibility constraints underlying clientelism. When electoral competition is intense and the probability of winning is relatively low, political parties have a stronger incentive to exert clientelistic pressure on non-elite voters to secure their support. As the probability of electoral victory increases, the party's electoral position becomes more secure and the need to intensive clientelistic enforcement diminishes. Thus, clientelistic effort and electoral security are substitutes: greater electoral security allows the party to reduce the resources devoted to maintaining non-elite support.

The welfare weights provide a complementary interpretation of this adjustment. Since $\Theta_E = 1$,

the welfare weight of elites is determined by their mean swing propensity and remains constant as γ changes. This reflects the facts hat elite voters respond primarily through the transfer channel: because their voting decision is non-instrumental, clientelistic pressure has no effect on their behaviour, making private the relevant instrument through with the party can secure their support. A higher political welfare weight on elites therefore allows the party to use private transfers to attract and retain their votes. Non-Elite voters, in contrast, respond instrumentally and can be influenced through clientelistic pressure. When electoral competition is high, the party can therefore rely more heavily on clientelistic enforcement to secure non-elite support rather than increasing their welfare weight.

As electoral victory becomes more secure, the party can reduce clientelistic pressure while placing relatively greater weight on non-elite voters. This explains why the welfare weight ratio δ_E/δ_{NE} declines with γ. As electoral security increases, the relative importance of elite support falls, but does not disappear. Indeed, the ratio remains above unity throughout the range of γ, indicating that elites continue to receive a higher political welfare weight than non-elites even when the party's electoral position is relatively secure. The result therefore suggest that political parties use different instruments to secure support from the two groups: private transfers are relatively more important for securing elite votes, while clientelistic pressure provides an alternative mechanism for securing non-elite support, particularly under greater electoral competition.

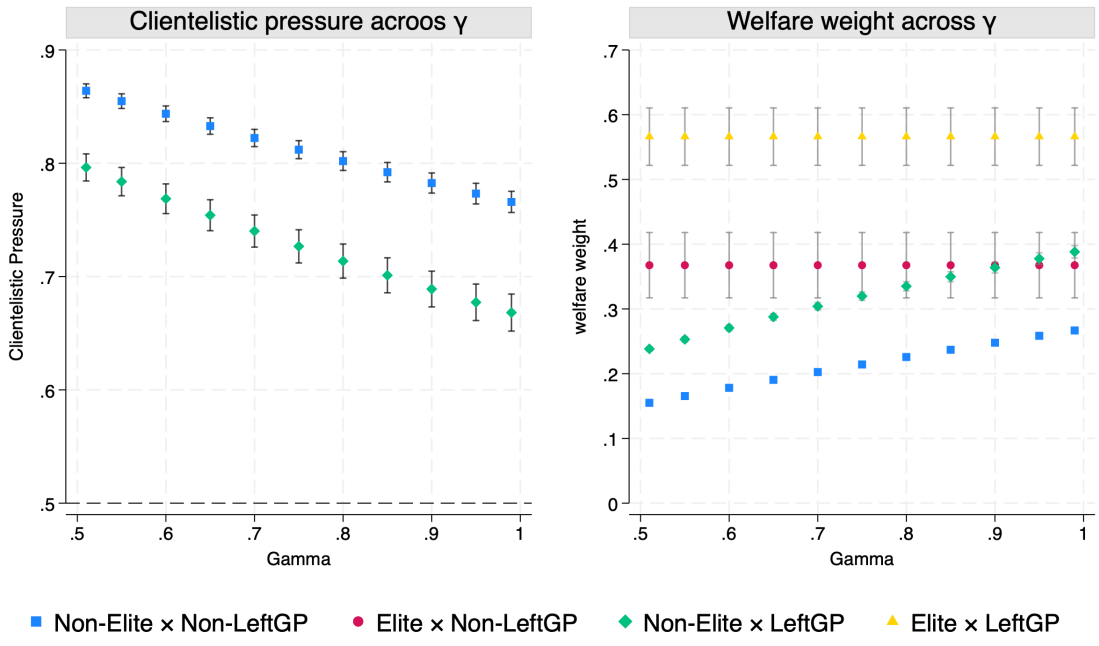


Figure 10: Caption

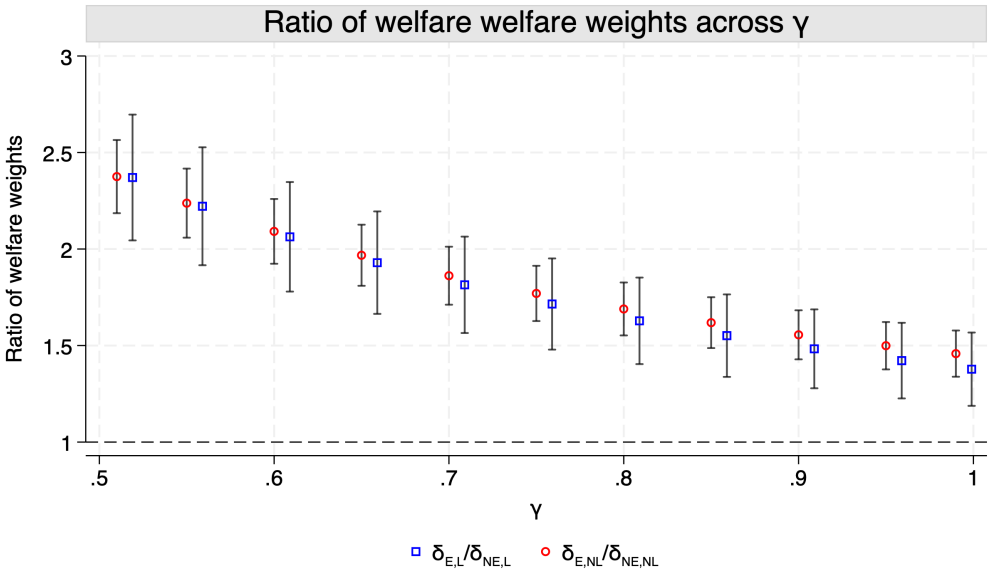


Figure 11: Caption

10.2 Fixed point iteration

Two step GMM ; **Cluster** = 88 **village level, Converged = True**							
Left Region				Non-Left Region			
recovered *Clientelistic parameter*		*recovered welfare wt.*		recovered *Clientelistic parameter*		ratio of *welfare wt.*	
parameter	mean	parameter	mean	parameter	mean	parameter	mean
$\gamma_L^* = 0.7813$				$\gamma_{NL}^* = 0.6758$			
$z_{NE,L}$	0.7185***	$\delta_{NE,L}$	0.3294	$z_{NE,NL}$	0.8273***	$\delta_{NE,NL}$	0.1967
	(0.0076)		(0.0034)		(0.0038)		(0.0009)
$z_{E,L}$	0.0000***	$\delta_{E,L}$	0.5662	$z_{E,NL}$	0.0000	$\delta_{E,NL}$	0.3676
	(0.0000)		(0.0226)		(0.0000)		(0.0257)
ratio		1.7183***		ratio		1.8677***	
		(0.0182)				(0.0085)	
$***p < 0.01, **p < 0.05, *p < 0.10.$							

Table 13: Result from contraction mapping

The sensitivity leaves γ undetermined within admissible range. To resolve this, the winning probability is determined endogenously through fixed contraction mapping, which selects the value of each γ in each region consistent with the model's equilibrium fixed point condition. Table 13 reports the estimates evaluated at these self-consistent winning probabilities.

The contraction mapping resolves the indeterminacy in γ by requiring the probability of winning used by voters to evaluate the credibility of clientelistic promises to coincide with the probability generated by the equilibrium. The resulting equilibrium probabilities $\gamma_L^* = 0.6758$ and $\gamma_{NL}^* = 0.7813$. By optimally adjusting transfers and clientelistic pressure, political parties can secure

electoral victory even as electoral competition intensifies. The result also indicate that parties rely on different mechanisms to secure support across voter groups. Clientelistic pressure is positive and statistically significant for non-elites, while it is essentially zero for elites in both regions. This suggests that non-elite support is secured through clientelistic pressure, whereas elite support is secured primarily through the allocation of private transfers.

The recovered welfare weights further indicate that political parties place substantially higher weight on elites than on non-elites. The ratio of the elite to non-elite welfare welfare weight is approximately 1.7183 in Left region and 1.8677 in Non-Left regions, and both ratios are statistically significant. Implies, having higher ex-ante endowment allows the voter choose strictly based on the policy choice made by the political party and not rely on clientelistic threats from political parties. From the political party's' perspective, clientelistic threats will not work on the elite, thus the only way to capture their vote is by choosing policy that significantly increase elites endowments. On the other hand, transfers and clientelistic pressure act as a substitutes for the non-elites. This creates a political equilibrium in which electoral competition does not necessarily translate into more equal resource allocation, parties can remain electorally viable while maintaining a substantially higher political welfare weight.

11 Comparison

The estimates from the two specifications are broadly consistent in their main implications, but the endogenous specification provides a much sharper characterisation of the underlying electoral mechanism. In the first specification, both θ and clientelistic pressure are treated as exogenous objectives. Consequently, the estimated welfare vary over a range of values as these parameters are changed. In second specification, these components are instead determined within the model: the type of voter behaviour depends on initial endowment, while clientelistic pressure is allowed to respond to the same underlying household characteristics. This links the estimated parameters to the economic environment rather than treating them as independent calibration choices.

This difference is reflected most clearly in the private goods policy distribution preference. The first specification produces relatively wide ranges, particularly for the relative weight placed on elites. In contrast, the endogenous model yields a single ratio of 1.895 for non left GP and 1.747 for Left GPs, compared with the point identified ratios of 1.865 and 1.825, respectively. The estimates therefore remains close to the point identified values while being obtained

without fixing θ or the probability of clientelistic pressure externally. The sharper estimates are consequence of imposing more of the structure implied by the theoretical model on the estimation.

The endogenous specification also gives a clearer interpretation of the difference between elites and non-elites. Clientelistic pressure is estimated to be positive for non-elites but zero for elites in both regions. This is consistent with the proposition that clientelistic control is effective only when voters are economically disadvantaged. When voters have limited economic resources, the value of clientelistic benefits is sufficiently high for parties to influence their electoral behaviour through such exchanges. As the economic position of the electorate improves, the scope for sustaining clientelistic influence diminishes. The zero estimated clientelistic pressure on elites and the positive pressure on non-elites are consistent with this distinction: non-elites remain responsive to clientelistic incentives, where as more affluent voters are less susceptible to them.

At the same time, elites receive a higher political welfare weight: The estimated ratio δ_E/δ_{NE} is greater than one in both regions. The model therefore separates two channels through which parties can secure electoral support. Non-elites can be reached through clientelistic pressure, while elite support is reflected primarily in the allocation of welfare weight and transfers. This distinction is obscured when clientelistic pressure is treated simply as an exogenous probability.

The regional estimates are also consistent with this interpretation. The non-Left regions have a lower estimated probability of electoral victory ($\gamma^{NL} = 0.6643$) and higher clientelistic pressure on non-elites ($z_{NE,NL} = 0.8298$) than the Left regions ($\gamma^{L} = 0.7633$ and $z_{NE,L} = 0.7231$). This, the model does not merely recover different parameter values. The estimated parameters move together in the direction implied by the theory. Greater electoral uncertainty is associated with greater reliance on clientelistic pressure, while the resulting allocation continues to place greater political weight on elites.

Taken together the comparison suggests that the endogenous specification provides a more internally consistent representation of the electoral process. Rather than asking how the estimated welfare weights change when θ or clientelistic pressure is assigned different values. The model uses the observed allocation of benefits and the equilibrium condition to determine these objects jointly. The resulting estimates are therefore more tightly tied to the mechanisms in the theoretical model, while retaining the central qualitative finding of the original specification: elite receive greater political weight than non-elites.

12 Counterfactual

The estimated welfare weights $\hat{\delta}_{g,r}$ summarise how the allocation of resources values each group. To see what they imply, I measure how a planner holding these weights would allocate a budget B, and compare the answer with an egalitarian benchmark in which every group receives the same weight ($\delta_{g,r} = 1$). In both cases the planner maximises $\sum_{g,r} \mu_{g,r}\delta_{g,r} \ln(\omega_{g,r} + q^*_{g,r})$ subject to the regional budget constraints. B is varied from 0 to 10 and hold the regional split.

With logarithmic utility the solution is available in closed form. Once both groups receive transfers, post-transfer resources satisfy

$$\frac{\omega_{r,E} + q^*_{r,E}}{\omega_{r,NE} + q^*_{r,NE}} = \frac{\delta_{r,E}}{\delta_{r,NE}}. \tag{27}$$

Relative resources therefore depend only on relative welfare weight, and not on the size of the budget or of the groups. Under the benchmark the ratio is one, the planner first brings the non-elite up to the elite's endowment and then raises both groups together. Figure 12 reports the results.

The estimated weights sustain a permanent elite premium. By (27), the political parties set elite preference to be 1.87 times that of non-elite in the Non-Left region and about 1.72 time in the left regions, at every budget level. A larger budget raises both groups but never closes the gap, which is labled as *structural inequality* (light orange shade). The premium is smaller in left regions, where the estimated weight are less tilted towards elites.

The Non-elite stop being priority early. Because the non-elite start with low endowment, both planners direct the first transfers to them. The egalitarian planner continues until the non-elite reach parity with the elite. Under the estimated weights, the elite begin to receive transfers once non-elites have reached only 53.5 per cent (non-left) and 58.2 per cent (left) of the elite's endowment.

The cost to the non-elite depend on how numerous the elite are. The dark shaded area, which is label as *structural violence*, is the shortfall of non-elite resources relative to the egalitarian benchmark. With the observed population shares (panel A) the shortfall is 13.4 percent, in the non-left and 12.4 percent in the left regions, while the elite receives 61.9 and 50.6 per cent more than under the benchmark. Favouring a small elite is thus cheap for the majority, large gains are concentrated on a few. When the population shares are swapped so that the elite are the

majority (Panel B), the same weight cost the non-elite 41.7 per cent in the non-left and 36.6 per cent in the Left, while each member of the elite gains only 9 percent. Measuring population shares within regions (Panels C and D) lowers the level of resources per person but leaves these proportions unchanged.

These counterfactuals hold the welfare weights, endowments and regional budget splint fixed, and abstract from behavioural responses to transfers. They should be read as an accounting of what the estimated weights imply for the distribution of resources, not a forcast of a policy reform.

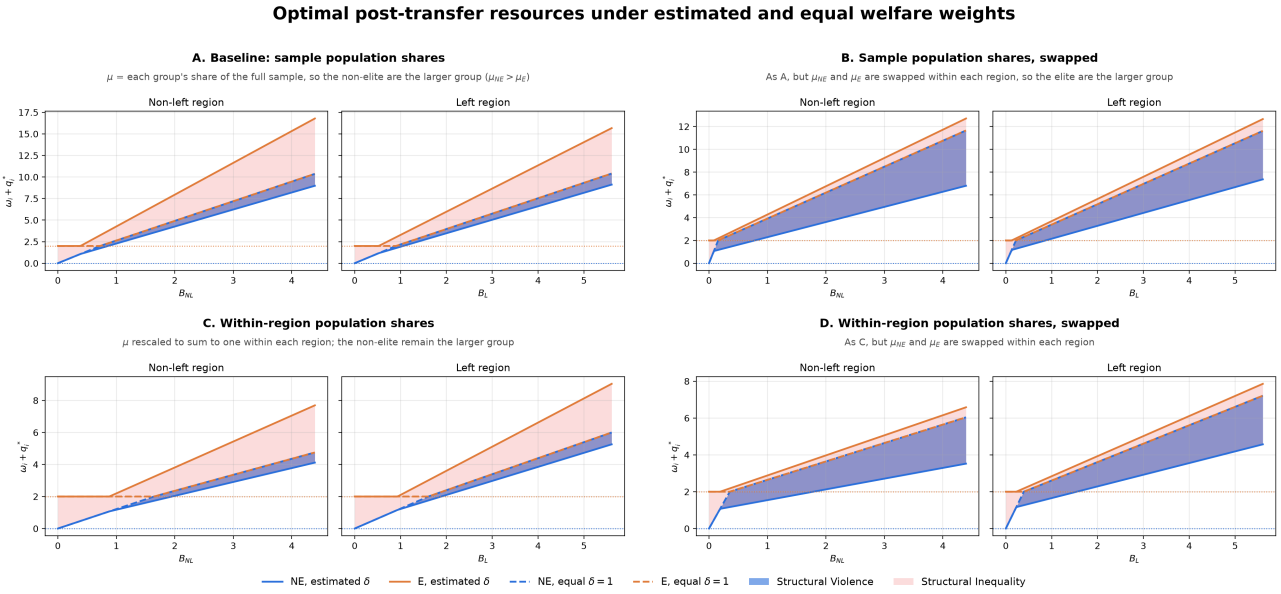


Figure 12: Counter Factual allocation under estimated and equal welfare weights

Notes: Each panel plots post transfer resources $\omega_i + q_i^*$ against the regional budget. Solid lines use the estimated welfare weights; dashed lines use equal weights. Dotted lines mark initial endowments. *Panel A* uses sample population shares; *Panel B* swaps the elite and non-elites shares with in each regions; *Panel C* and *Panel D* repeat *A* and *B* with shares normalised to one within each region.

13 Concluding Comments

I have presented an extension to the theory of clientelism and elite capture, which generates a number of novel results. To derive these results, the model endogenises the voter's choice function, allowing voters to self select into one of two types, instrumental and non-instrumental. Voter's type is determined by initial endowments. Poorer groups, motivated by immediate material benefits, choose to vote instrumentally, basing their decision on expected utility that reflects both benefits and clientelistic pressure. Wealthier groups vote non-instrumentally, reflecting a lesser reliance on clientelistic transfers. Clientelistic control is also endogenised: Political party chooses the level of control is exerts over group i.

The model yields three main proposition, which together characterise the equilibrium. First, in unequal society, both parties assign the same welfare weight to the elite and same welfare weight to the non-elite and probability of winning determined by the weight of the clientelistic pressure. Second, poorer groups optimally choose instrumental voting while wealthier groups behave non-instrumentally, and parties set clientelistic control as a function of voters' initial endowments. It is optimal for parties to exert pressure on the non-elite only, since such pressure has no effect on the elite. Third, where inequality is high, clientelism leads parties to assign a disproportionately higher welfare weight to the elite, amplifying inequality through the electoral process. When the component of elite capture is added to this framework, it is seen that clientelism and elite capture operate as complementary rather than opposing mechanism, jointly reinforcing unequal outcomes.

The structural estimation takes the clientelism side of this framework to household data from rural West Bengal. Because the parameters are estimated within the equilibrium of the model, they are consistent with the optimising behaviour of both voters and parties. Reduced form analysis cannot guarantee this behaviour. The estimated model is the used to generate out of sample predictions.

The first specification of model shows that political parties have regressive preference when it comes to the private goods distribution. In GPs governed by the Left alliance, these preferences become egalitarian or progressive for certain values of voter type parameter. Overall, the relative welfare weights estimated under the first specification are close to unity. There are therefore less regressive than then point identified estimates, which suggests that treating voter type as exogenous pulls the estimated preferences towards progressive preference.

The second specification makes both the voters' choice of type and the parties' choice of clientelistic pressure endogenous. voter choose their type after observing the parties' policy platforms. The rich choose to vote non-instrumentally, so that their vote depends only on policy. The poor choose to vote instrumentally, so their vote depends only on policy, on clientelistic pressure, and on their beliefs about which party will win.

This sorting determines each party's optimal strategy. To win elites votes, a party maximises transfers to the elite and exerts non clientelistic pressure on them. To win non-elite votes, it keeps their transfers just above the subsistence level and maximises the clientelistic pressure.

The estimation results show that political parties have regressive preferences in both regions, and that this finding is robust to the likelihood of wining. The preferences obtained from the

fixed point results are also close to the point identified estimates.

Two limitations should be noted. The estimation abstracts from elite capture, because the data contain no information on campaign finance or elite contributions. The complementarity between clientelism and capture is therefore a theoretical result that remains to be tested empirically. The estimates are alos specific to one Indian state over 2008 - 2011, and their relevance to other setting is an open question.

Interpreted through the lens of structural violence, the democratic process in an unequal society systematically reproduces and deepens existing disparity rather than mitigating then. This paper contributes to the literature by showing theoretically that clientelism and elite capture reinforce one another, and by providing estimates how strongly the clientelism mechanism operates in practice. Extending the estimation to include capture, as data on political finance become available, is a natural direction for future research.

A Appendix (Theory)

A.1 Maximising the utilitarian optimal allocation maximiser

Maximise the utilitarian optimal allocation subject to budget constraint which is equal to $\mu_c q_c + \mu_e q_e = B$

$$\mathcal{L} = \sum_{i \in \{c,e\}} \mu_i u(\omega_i + q_i) - \lambda(\mu_c q_c + \mu_e q_e - B)$$

FOCs:

- $\frac{\partial \mathcal{L}}{\partial q_c} \equiv \mu_c u'(\omega_c + q_c) - \lambda \mu_c = 0$
- $\frac{\partial \mathcal{L}}{\partial q_e} \equiv \mu_e u'(\omega_e + q_e) - \lambda \mu_e = 0$
- $\frac{\partial \mathcal{L}}{\partial \lambda} \equiv \mu_e q_e + \mu_c q_c - B = 0$

From FOC's:

$$\lambda = u'(\omega_c + q_c) = u'(\omega_e + q_e) \text{ and hence } \lambda > 0$$

The second order conditions are as follows:

- $\frac{\partial^2 \mathcal{L}}{\partial q_c^2} \equiv u''(\omega_c + q_c) < 0$
- $\frac{\partial^2 \mathcal{L}}{\partial q_e^2} \equiv u''(\omega_e + q_e) < 0$

$\implies u'$ is invertible.

$\therefore (u')^{-1}\lambda = \omega_c + q_c = \omega_e + q_e$

- $q_c = (u')^{-1}(\lambda) - \omega_c$
- $q_e = (u')^{-1}(\lambda) - \omega_e$

Substituting the values of q_c and q_e into budget constraint.

$$= \mu_e \left((u')^{-1}(\lambda) - \omega_e\right) + \mu_c \left((u')^{-1}(\lambda) - \omega_c\right) = B$$
$$= (u')^{-1}(\lambda) = B + \mu_e \omega_e + \mu_c \omega_c$$

Therefore, $q_c^* = B + \mu_e \omega_e + \mu_c \omega_c - \omega_c$ and $q_e^* = B + \mu_e \omega_e + \mu_c \omega_c - \omega_e$

From above we can conclude that $\{q_i\} > 0$

The above calculation proves that utilitarian welfare allocation is maximally progressive, i.e., Allocation of government benefits reduce inequality as much as possible. E.g., If $\omega_c = 0$, $\omega_e = 10$, $\mu_c = \mu_e = 0.5$, $B = 30$. The local government augments benefits to two groups such that final endowments of both groups are equal. $q_c^* = 35$ $q_e^* = 25$.

Maximising Inequality Budget Constraint

Let's maximise the utilitarian optimal allocation subject to budget constraint being equal $\mu_c q_c + \mu_e q_e \leq B$

$$\mathcal{L} = \sum_{i \in \{c,e\}} \mu_i u(\omega_i + q_i) - \lambda(\mu_c q_c + \mu_e q_e - B)$$

FOCs:

- $\frac{\partial \mathcal{L}}{\partial q_c} \equiv \mu_c u'(\omega_c + q_c) - \lambda \mu_c = 0$
- $\frac{\partial \mathcal{L}}{\partial q_e} \equiv \mu_e u'(\omega_e + q_e) - \lambda \mu_e = 0$

Suppose we maximise the utility function subject to the inequality of we get complementary slack condition:

$$\lambda(\mu_c q_c + \mu_e q_e - B) = 0$$

There are two possible solutions to this slackness condition:

- $\lambda = 0 \implies \mu_c q_c + \mu_e q_e < B$ Interior solution or $\mu_c q_c + \mu_e q_e = B$ Boundary solution
  $0 = u'(\omega_c + q_c) = u'(\omega_e + q_e)$ marginal utility remains constant.
- $\lambda \neq 0$
  - $\lambda > 0 \implies \mu_c q_c + \mu_e q_e = B$ Boundary Solution.
    $0 > u'(\omega_c + q_c) = u'(\omega_e + q_e)$ marginal utility is positive thus $\{q_i\} > 0$
  - $\lambda < 0 \implies \mu_c q_c + \mu_e q_e = B$ Boundary Solution.
    Marginal utility is negative. Therefore not a solution.

The second order conditions are as follows:

- $\frac{\partial^2 \mathcal{L}}{\partial q_c^2} \equiv u''(\omega_c + q_c) < 0$
- $\frac{\partial^2 \mathcal{L}}{\partial q_e^2} \equiv u''(\omega_e + q_e) < 0$

$\implies u'$ is invertible.

$q_e = (u')^{-1}(\lambda) - \omega_e$

$$q_c = (u')^{-1}(\lambda) - \omega_c$$

$$q_e = (u')^{-1}(\lambda) - \omega_e$$

Substituting the values of q_c and q_e into budget constraint.

$$-\mu_e\left((u')^{-1}(\lambda) - \omega_e\right) + \mu_c\left((u')^{-1}(\lambda) - \omega_c\right) = B$$
$$-(u')^{-1}(\lambda) = B + \mu_e\omega_e + \mu_c\omega_c$$

Therefore $q_c^* = B + \mu_e\omega_c + \mu_e\omega_e - \omega_c$ and $q_e^* = B + \mu_c\omega_c + \mu_e\omega_e - \omega_e$

From above we can conclude that $\{q_i\} > 0$

A.2 Maximising the quasi-utilitarian optimal allocation maximiser

Maximisation problem :

$$\sum_{i\in\{e,c\}} \mu_i\delta_i u(\omega_i + q_i)$$

subject to budget constraint: $\mu_e q_e + \mu_c q_c \leq B$ and $\{q_i^p > 0\}$

Maximisation

$$\mathcal{L} = \sum_{i\in\{c,e\}} \mu_i\delta_i u(\omega_i + q_i) - \lambda(\mu_c q_c + \mu_e q_e - B)$$

FOCs:

- $\frac{\partial\mathcal{L}}{\partial q_c} \equiv \mu_c\delta_c u'(\omega_c + q_c) - \lambda\mu_c = 0$
- $\frac{\partial\mathcal{L}}{\partial q_e} \equiv \mu_e\delta_c u'(\omega_e + q_e) - \lambda\mu_e = 0$

Generalised FOC:

$$\frac{\partial\mathcal{L}}{\partial q_i} \equiv \mu_i\delta_i u'(\omega_i + q_i) - \lambda\mu_i = 0 \tag{A.1}$$

Suppose we maximize the utility function subject to the inequality of we get complementary slack condition:

$$\lambda(\mu_c q_c + \mu_e q_e - B) = 0$$

There are two possible solutions to this slackness condition:

- $\lambda = 0 \implies \mu_c q_c + \mu_e q_e < B$ Interior solution. OR $\mu_c q_c + \mu_e q_e = B$ Boundary solution.
  $0 = u'(\omega_c + q_c) = u'(\omega_e + q_e)$ marginal utility remains constant.
- $\lambda \neq 0$
  - $\lambda > 0 \implies \mu_c q_c + \mu_e q_e = B$ Boundary Solution.
    $0 > u'(\omega_c + q_c) = u'(\omega_e + q_e)$ marginal utility is positive thus $\{q_i\} > 0$

Utilities are concave and additive separable.

$$(u')^{-1}\frac{\lambda}{\delta_i} = \omega_c + q_c = \omega_e + q_e$$

At the optimal solution, we get the condition:

$$\frac{u'(\omega_e + q_e)}{u'(\omega_c + q_c)} = \frac{\delta_c}{\delta_e}$$

$$q_c^* = (u')^{-1}\frac{\lambda}{\delta_c} - \omega_c$$

$$q_e^* = (u')^{-1}\frac{\lambda}{\delta_c} - \omega_e$$

substituting the values of q_c^* and q_e^* into the budget constraint:

$$\mu_e((u')^{-1}\frac{\lambda}{\delta_e} - \omega_e) + \mu_c((u')^{-1}\frac{\lambda}{\delta_c} - \omega_c) = B$$

$$\implies \mu_e((u')^{-1}\frac{\lambda}{\delta_e}) + \mu_c((u')^{-1}\frac{\lambda}{\delta_c}) = B + \mu_c\omega_c + \mu_e\omega_e$$

The above equation must solve for λ.

If we use **log utility** then:

$$q_c^* = \frac{\delta_c(B + \mu_c\omega_c + \mu_e\omega_e)}{\delta_c\mu_c + \delta_e\mu_e} - \omega_c$$
$$q_e^* = \frac{\delta_e(B + \mu_c\omega_c + \mu_e\omega_e)}{\delta_c\mu_c + \delta_e\mu_e} - \omega_e$$

looking at the quasi utilitarian optimal allocation function, we compare δ_i and q_i^*:

- If, $\delta_c = \delta_e \implies$ over all consumption is same.
- If, $\delta_c > \delta_e \to q_c^* > q_e^* \implies$ Overall consumption is more for citizen than elite, endowment

is more for c.

- If, $\delta_c < \delta_e \rightarrow q_c^* < q_e^* \implies$ endowment for elite is more and overall consumption is more for the elite, increases the inequality.

How does change in inequality of initial endowment affect the final consumption? How does change in welfare weight affect the final consumption? Table A.1 presents a numerical examples answering the above questions.

		case 1	*case 2*	*case 3*
	Initial Conditions			
Utility			$u(.) = \log(.)$	
Poor hh endowment	(ω_c)	0	0	0
Rich hh endowment	(ω_e)	**20**	**40**	**20**
Welfare weights	(δ_c, δ_e)	(1,2)	(1,2)	**(1,4)**
Budget	(B)	30	30	30
Population shares	(μ_c, μ_e)	(0.5,0.5)	(0.5,0.5)	(0.5,0.5)
	Equilibrium Transfers			
Transfer to poor	(q_c^*)	26.67	33.33	16
Transfer to elite	(q_e^*)	33.33	26.67	44
	Final consumption			
Poor consumption per hh	$(\omega_c + q_c^*)$	26.67	33.33	16
Rich consumption per hh	$(\omega_e + q_e^*)$	53.33	66.66	64
Consumption Gap		26.66	33.33	48

Notes: case 1 and *case 2* differ only in the elite household's initial endowment. All other parameters are held constant. Equilibrium transfers are obtained from the planner's allocation problem subject to the budget constraint. *case 1* and *case 3* differ only in the welfare weight of the elite group. All other parameters are held constant. Equilibrium transfers are obtained from the planner's problem subject to the budget constraint.

Table A.1: Change in consumption gap

A.3 Calculating Vote Share

The uniform distribution centred around l_i is given by:

$$\epsilon_i \sim U\left(l_i - \frac{1}{2\sigma_i}, l_i + \frac{1}{2\sigma_i}\right)$$

The CDF is:

$$F_{\epsilon_i}(x) = \begin{cases} 0, & x < l_i - \frac{1}{2\sigma_i} \\ \frac{x - \left(l_i - \frac{1}{2\sigma_i}\right)}{\frac{1}{\sigma_i}}, & l_i - \frac{1}{2\sigma_i} < x < l_i + \frac{1}{2\sigma_i} \\ 1, & x > l_i + \frac{1}{2\sigma_i} \end{cases}$$

step 1: Setting vote probability.

Now we identify the probability of the vote preference, A voter in group i votes for party L if their net utility from doing so is positive:

$$Pr(\Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R] + \varepsilon_i > 0)$$

d is defined as the deterministic utility:

$$d \equiv \Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R]$$

The vote probability becomes:

$$\Pr(\varepsilon_i > -d) = 1 - \Pr(\varepsilon_i \leq -d) = 1 - F_{\varepsilon_i}(-d)$$

Step 2: Applying the CDF

Substituting $x = -d$ into the CDF (assuming '$-d$' lies in the support):

$$F_{\epsilon_i}(-d) = \frac{-d - \left(l_i - \frac{1}{2\sigma_i}\right)}{\frac{1}{\sigma_i}}$$

$$1 - F_{\epsilon_i}(-d) = 1 - \frac{-d - \left(l_i - \frac{1}{2\sigma_i}\right)}{\frac{1}{\sigma_i}}$$

Step 3: Simplifying the fraction:

$$\begin{aligned} &= 1 - \frac{-d - \left(l_i - \frac{1}{2\sigma_i}\right)}{\frac{1}{\sigma_i}} \\ &= 1 - \sigma_i \left(-d - l_i + \frac{1}{2\sigma_i}\right) \\ &= \frac{1}{2} + \sigma_i l_i + \sigma_i d \end{aligned}$$

Step 4: Substituting back d

$$= \frac{1}{2} + \sigma_i l_i + \sigma_i \left\{\Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R]\right\}$$

step 5: Aggregating Over the Population:

The vote share of party L is the population-weighted average of each group's vote probability,

where μ_i is the population share of group i with $\sum_i \mu_i = 1$:

$$S_L(\gamma^L; q^L, q^R) \equiv \sum_i \mu_i \Bigg((\frac{1}{2} + l_i\sigma_i + \\ \sigma_i \ \{\Theta_i \left[u(\omega_i + q_i^L) - u(\omega_i + q_i^R)\right] + \\ (1-\Theta_i)(\gamma_L z_i^L \left[u(\omega_i + q_i^L) - u(\omega_i)\right] - (1-\gamma_L) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)]\} \Bigg)$$

$$S_L(\gamma^L; q^L, q^R) \equiv \frac{1}{2} + \sum_i \mu_i l_i \sigma_i + \sum_i \mu_i \sigma_i \\ \{\Theta_i \left[u(\omega_i + q_i^L) - u(\omega_i + q_i^R)\right] + \\ (1-\Theta_i)(\gamma_L z_i^L \left[u(\omega_i + q_i^L) - u(\omega_i)\right] - (1-\gamma_L) z_i^R [u(\omega_i + q_i^R) - u(\omega_i)]\}$$

A.4 Calculating the welfare weight δ_i

We first start with equation 10, find the the first order derivative with respect to q_i^L, rearrange and reach equation 12.

Probability of winning is given by:

$$\gamma_L = \phi(S_L(\gamma_L; \gamma_L(q^L, q^R)))$$

Maximising the above equation wrt (q_i^p) and using the implicit function theorem.

$$\frac{\partial \gamma_L}{\partial q_i^L} = \phi'(S_L) \left(\frac{\partial S_L(\gamma_L; \gamma_L(q^L, q^R))}{\partial q_i^L} \right)^{7}$$

$$\frac{\partial \gamma_L}{\partial q_i^L} = \phi'(S_L) \left(\frac{\partial S_L}{\partial q_i^L} \frac{\partial q_i^L}{\partial q_i^L} + \frac{\partial S_L}{\partial \gamma_L} \frac{\partial \gamma_L}{\partial q_i^L} \right)$$

$$\frac{\partial \gamma_L}{\partial q_i^L} = \phi'(S_L) \left(\frac{\partial S_L}{\partial q_i^L} 1 + \frac{\partial S_L}{\partial \gamma_L} \frac{\partial \gamma_L}{\partial q_i^L} \right)$$

$$\frac{\partial \gamma_L}{\partial q_i^L} \left(1 - \phi'(S_L) \frac{\partial S_L}{\partial \gamma_L} \right) = \phi'(S_L) \left(\frac{\partial S_L}{\partial q_i^L} 1 \right)$$

$$\frac{\partial \gamma_L}{\partial q_i^L} = \frac{\phi'(S_L) \left(\frac{\partial S_L}{\partial q_i^L} \right)}{\left(1 - \phi'(S_L) \frac{\partial S_L}{\partial \gamma_L} \right)}$$

$$\frac{\partial \gamma_L}{\partial q_i^L} = \frac{\phi'(S_L)\left(\mu_i\sigma_i[\Theta_i + (1-\Theta_i)\gamma_L z_i^L]u'(\omega_i + q_i^L)\right)}{1 - \phi'(S_L)(1-\Theta_i)\sum_i \mu_i\sigma_i[z_i^L(u(\omega_i + q_i^L) - u(\omega_i)) + z_i^R(u(\omega_i + q_i^R) - u(\omega_i))]}$$

$$\frac{\partial \gamma_L}{\partial q_i^L} = J^L\left[\mu_i\sigma_i[\Theta_i + (1-\Theta_i)\gamma_L z_i^L]u'(\omega_i + q_i^L)\right]$$

$$\frac{\partial \gamma_L}{\partial q_i^L} \cdot \frac{1}{J^L} = \left[\mu_i\sigma_i[\Theta_i + (1-\Theta_i)\gamma_L z_i^L]u'(\omega_i + q_i^L)\right] \tag{A.2}$$

$$J^L = \frac{\phi'(S_L)}{1 - \phi'(S_L)(1-\Theta_i)\sum_i \mu_i\sigma_i[z_i^L(u(\omega_i + q_i^L) - u(\omega_i)) + z_i^R(u(\omega_i + q_i^R) - u(\omega_i))]}$$

Equation 9 ensures that denominator in J^L is positive. Comparing A.1 with A.2 we obtain:

$$\lambda\mu_i = \mu_i\ \delta_i\ u'(\omega_i + q_i)$$

$$\frac{\partial \gamma_L}{\partial q_i^L} \cdot \frac{1}{J^L} = \left[\mu_i\sigma_i[\Theta_i + (1-\Theta_i)\gamma_L z_i^L]u'(\omega_i + q_i^L)\right]$$

$$\delta_i = \sigma_i[\Theta_i + (1-\Theta_i)\gamma_L z_i^L]$$

The quasi utilitarian welfare function is as follows:

$$\sum_{i\in\{e,c\}} \mu_i[\Theta_i + (1-\Theta_i)\gamma_L z_i^L]u(\omega_i + q_i^L)$$

A.5 Winning probability and equilibrium fixed point

A.6 *Proof* of Proposition 1

Proof :

Each party $p \in \{L, R\}$ solves

$$\max_{\{q_i^p\}_{i\in\{e,c\}}} \sum_{i\in\{e,c\}} \mu_i\delta_i^p u(\omega_i + q_i^p)$$

subject to the budget constraint and non-negativity constraint. suppose

$$\delta_e^L = \delta_e^R = \delta_e, \quad \delta_c^L = \delta_c^R = \delta_c, \quad \delta_e \neq \delta_c$$

Both parties face identical optimization problems. Since the objective functions and constraints

[7] $\frac{\partial S_L(\gamma_L;\gamma_L(q^L,q^R))}{\partial q_i^L} = \frac{\partial S_L}{\partial q_i^L}\frac{\partial q_i^L}{\partial q_i^L} + \frac{\partial S_L}{\partial \gamma_L}\frac{\partial \gamma_L}{\partial q_i^L}$

are the same, the equilibrium policy platforms much coincide:

$$q_i^{L*} = q_i^{R*} = q_i, \qquad i \in \{e, c\}.$$

Therefore, regardless of which party wins the elections, the implemented transfer for group i is q_i^*. The ex-post utility of each group is thus

$$u_i^* = u(\omega_i + q_i^*),$$

which is independent of the identity of winning party. Hence, electoral outcomes do not affect the final utility of either elites or non-elites.

However, equality of utilities across groups does not follow. Since δ_e and δ_c may differ, the optimal transfer allocations q_e^* and q_c^* generally differ. Consequently,

$$u(\omega_e + q_e^*) \neq u(\omega_c + q_c^*),$$

in general. Thus, utilities are invariant to electoral outcomes within each group, but need not be equalized across groups. □

A.7 *Proof* of lemma 1

A.8 *Proof* of Proposition 2.1

Proof :

We prove this proposition by using the intermediate value theorem. We know that voting behaviour is given by:

$$\Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R] + \varepsilon_i > 0$$

voters is choice is derived by maximising the voting behaviour w.r.t. Θ_i

$$\max_{0 \leq \Theta_i \leq 1} \left\{ \Theta_i[N_i^L - N_i^R] + (1 - \Theta_i)[I_i^L - I_i^R] + \varepsilon_i \right\}$$

Since $u(\cdot)$ is continuous and strictly concave ($u'' < 0$), both $N_i^L - N_i^R$ and $I_i^L - I_i^R$ are continuous in ω_i. Moreover, as $\omega_i \to 0$, the utility gain from clientelistic transfers is large relative to

the policy differential, implying that $N_i^L - N_i^R < I_i^L - I_i^R$, the voters behave instrumentally. In contrast, as $\omega_i \to \infty$, the Inada conditions implies that the marginal utility of transfers approaches zero, so that $N_i^L - N_i^R > I_i^L - I_i^R$, and voter's behaviour becomes instrumental.

By *Intermediate Value Theorem*, there exists $\omega_i^* \in (0, \infty)$ such that the voter is indifferent:

$$N_i^L - N_i^R|_{\omega_i^*} = I_i^L - I_i^R|_{\omega_i^*}$$

Wealth based voting partition is given by:

$$\begin{cases} \omega_i < \omega_i^*, & N_i^L - N_i^R < I_i^L - I_i^R \implies \Theta_i = 0 \text{ (Instrumental)} \\ \omega_i = \omega_i^*, & N_i^L - N_i^R = I_i^L - I_i^R \implies \Theta_i \in (0,1) \\ \omega_i > \omega_i^*, & N_i^L - N_i^R > I_i^L - I_i^R \implies \Theta_i = 1 \text{ (Non Instrumental)} \end{cases}$$

□

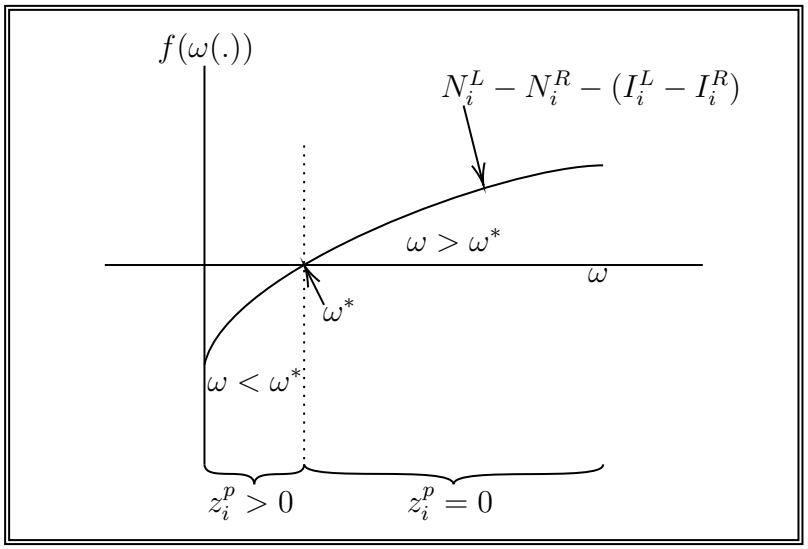


Figure A. 1: concave function of voting behaviour.

A.9 *Proof* of Proposition 2.2

Proof :

To prove the proposition, first I characterise the total clientelistic environment facing he voter. Define:

$$\Delta u_i^L(\omega_i) \equiv u(\omega_i + q_i^L) - u(\omega_i), \qquad \Delta u_i^R(\omega_i) \equiv u(\omega_i + q_i^R) - u(\omega_i)$$

As the utility gains from each party's clientelistic transfer relative to the voter's base endow-

ment. The net clientelistic pressure is then defined as the total weighted utility gains both parties can deliver through clientelistic control:

$$\mathcal{Z}(\omega_i) \equiv \gamma_L z_i^L(\omega_i)\,\Delta u_i^L(\omega_i) + (1-\gamma_L) z_i^R(\omega_i)\,\Delta u_i^R(\omega_i)$$

where $z_i^L(\omega_i), z_i^R(\omega_i) \in [0,1]$ are the endogenous clientelistic pressure parameters chosen by the left and the right party respectively. Each party independently contributes to the total clientelistic pull on the voter.

A voter chooses to be Instrumental ($\Theta_i = 0$) when $N_i^L - N_i^R < I_i^L - I_i^R$, i.e.,

$$u(\omega_i + q_i^L) - u(\omega_i + q_i^R) < \gamma_L\ z_i^L\ \Delta u_i^L - (1-\gamma_L)\ z_i^R\ \Delta u_i^R \tag{A.3}$$

The right hand side is the *partisan clientelistic advantage* to the left party. Rewriting in terms of $\mathcal{Z}(\omega_i)$:

$$\begin{aligned} \gamma_L z_i^L\,\Delta u_i^L - (1-\gamma_L) z_i^R\,\Delta u_i^R \quad &= \mathcal{Z}(\omega_i) - 2(1-\gamma_L) z_i^R \Delta u_i^R(\omega_i) \\ \underbrace{\gamma_L z_i^L\,\Delta u_i^L - (1-\gamma_L) z_i^R\,\Delta u_i^R}_{I_i^L - I_i^R} \quad &< \mathcal{Z}(\omega_i) \end{aligned} \tag{A.4}$$

substituting the above inequality from equation A.3 in equation A.4:

$$\begin{aligned} \underbrace{u(\omega_i + q_i^L) - u(\omega_i + q_i^R)}_{\mathcal{Z}^*(\omega_i) \equiv N_i^L - N_i^R} \quad &< \quad \gamma_L z_i^L\,\Delta u_i^L - (1-\gamma_L) z_i^R\,\Delta u_i^R \quad < \mathcal{Z}(\omega_i) \\ \mathcal{Z}^*(\omega_i) &< \mathcal{Z}(\omega_i) \end{aligned} \tag{A.5}$$

Equation A.5 is the necessary condition for clientelism to be effective. If the total clientelistic pressure $\mathcal{Z}(\omega_i)$ falls below the threshold $\mathcal{Z}^*(\omega_i)$, then no feasible choice of $z_i^L, z_i^R \in [0,1]$ can induce instrumental voting, and political parties fail to exert clientelism.

To understand the behaviour of the threshold function $\mathcal{Z}^*(\omega_i)$, differentiating the function w.r.t ω_i:

$$\frac{\partial \mathcal{Z}^*(\omega_i)}{\partial \omega_i} = u'(\omega_i + q_i^L) - u'(\omega_i + q_i^R) < 0$$

since $u'' < 0$ and $q_i^L > q_i^R$. So $\mathcal{Z}^*(\omega_i)$ is strictly decreasing in ω_i, but remains strictly positive for all finite ω_i. From above we identify the behaviour of $\mathcal{Z}(\omega_i)$ by using the Inada condition

and concavity of $u(\cdot)$. For any fixed transfers $q_i^L, q_i^R > 0$:

$$\Delta u_i^L(\omega_i) \to 0 \quad \text{and} \quad \Delta u_i^R(\omega_i) \to 0 \quad \text{as} \quad \omega_i \to \infty.$$

Since wealthy voters have negligible marginal utility from any fixed transfer. Therefore, regardless of the values of z_i^L and z_i^R: $\mathcal{Z}(\omega_i) \to 0 \quad \text{as} \quad \omega_i \to \infty$.

Conversely, at low endowment $\omega_i \approx 0$, transfers yield large utility gains so $\mathcal{Z}(\omega_i)$ is large. Since poor voters have significant marginal utility from any fixed transfer, therefore, for any positive values of z_i^L and z_i^R: $\mathcal{Z}(\omega_i) > 0 \quad \text{as} \quad \omega_i \to 0$.

We can identify the upper bound of $\mathcal{Z}(\omega_i)$

$$\mathcal{Z}_i(\omega_i) \leq \bar{\mathcal{Z}}(\omega_i) \equiv \gamma_L \, \Delta u_i^L(\omega_i) \times \max\{z_i^L\} + (1 - \gamma_L) \, \Delta u_i^R(\omega_i) \times \max\{z_i^R\}$$

where $\bar{\mathcal{Z}}(\omega_i)$ is strictly decreasing and converges to zero. Using the *Intermediate Value Theorem*, the boundary condition can be identified. First, Define the gap function $G(\omega_i) \equiv \mathcal{Z}(\omega_i) - \mathcal{Z}^*(\omega_i)$, then:

- As $\omega_i \to 0 : \mathcal{Z}(\omega_i)$ is large and $\mathcal{Z}^*(\omega_i)$ is finite, so $G(\omega_i) > 0 \implies$ clientelism is *feasible*.
- as $\omega_i \to \infty : \mathcal{Z}(\omega_i) \to 0$ and $\mathcal{Z}^*(\omega_i) > 0$, so $G(\omega_i) < 0 \implies$ clientelism is *infeasible*.

since $G(\omega_i)$ is continuous, the *Intermediate Value Theorem* there exists ω_i^{**} such that $G(\omega_i^{**}) = 0$, i.e.:

$$\mathcal{Z}(\omega_i^{**}) = \mathcal{Z}^*(\omega_i^{**})$$

This gives us the boundary condition:

$$\boxed{\gamma_L z_i^L [u(\omega_i^{**} + q_i^L) - u(\omega_i^{**})] + (1 - \gamma_L) z_i^R [u(\omega_i^{**} + q_i^R) - u(\omega_i^{**})] = u(\omega_i^{**} + q_i^L) - u(\omega_i^{**} + q_i^R)}$$

Wealth based clientelistic partition is given by:

$$\begin{cases} \omega_i < \omega_i^{**} : & \mathcal{Z}(\omega_i) > \mathcal{Z}^*(\omega_i) \implies \text{clientelism feasible} \\ \omega_i = \omega_i^{**} : & \mathcal{Z}(\omega_i) = \mathcal{Z}^*(\omega_i) \implies \text{boundary} \\ \omega_i > \omega_i^{**} : & \mathcal{Z}(\omega_i) < \mathcal{Z}^*(\omega_i) \implies \text{clientelism infeasible} \end{cases}$$

For any $\omega_i > \omega_i^{**}$, total clientelistic pressure falls below the minimum threshold $\mathcal{Z}_i^*(\omega_i)$, and no feasible assignment of z_i^L, z_i^R can sustain instrumental voting. □

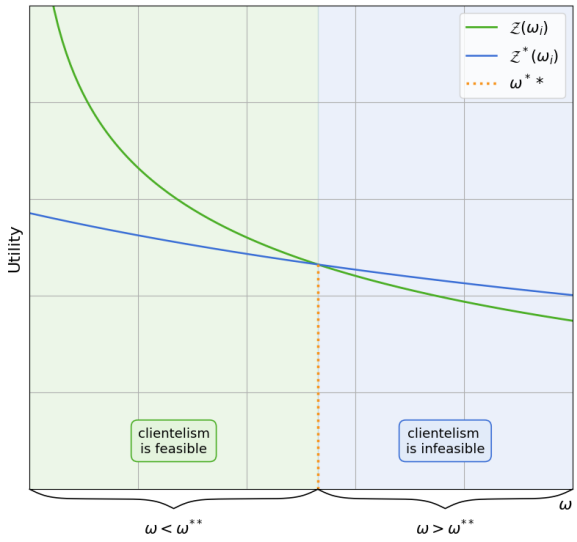


Figure A. 2: Clientelistic feasibility

A.10 *Proof* of Proposition 3

B Appendix : Structural Estimation

B.1 Derivations and Estimation Details

B.1.1 First-order conditions

The Lagrangian of the quasi-utilitarian problem is

$$\mathcal{L} = \sum_r \sum_g \mu_{g,r}\, \delta_{g,r}\, u(\omega_{g,r} + q_{g,r}) + \sum_{r \in \{L,NL\}} \lambda_r \Big[\sum_{g \in \{E,NE\}} \mu_{g,r}\, q_{g,r} - B_r \Big].$$

The first-order condition for $q_{g,r}$ is

$$\mu_{g,r}\, \delta_{g,r}\, u'(\omega_{g,r} + q_{g,r}) - \lambda_r\, \mu_{g,r} = 0 \quad \Longrightarrow \quad \delta_{g,r} = \frac{\lambda_r}{u'(\omega_{g,r} + q_{g,r})}.$$

Taking the ratio for $g = E$ and $g = NE$ within region r eliminates λ_r and yields (19). Under log utility, $u'(c) = 1/c$.

B.1.2 Deriving the estimating equation

Equating the first-order condition with the equilibrium weight (20) and using $u'(c) = 1/c$ gives

$$\lambda_r \left(\omega_{g,r} + q_{g,r}\right) = \bar{\sigma}_{g,r} \left[\Theta + (1 - \Theta)\, \gamma\, z_{g,r}\right].$$

Taking logs and rearranging,

$$\underbrace{\ln \lambda_r + \ln(\omega_{g,r} + q_{g,r}) - \ln \bar{\sigma}_{g,r}}_{\tilde{y}_{g,r}} = \ln \left[\Theta + (1 - \Theta)\, \gamma\, z_{g,r}\right].$$

Parametrising $z_{g,r} = \exp(x'b)$ log-linearly and adding a disturbance gives (21). The coefficients b are therefore log relative clientelistic pressures.

B.1.3 `GMM` identification

The disturbance ε_i captures measurement and implementation noise. The instruments Z_i are the group indicators plus a constant. Group traits are predetermined: they are fixed before ε_i is realised and do not respond to it. Hence $\mathbb{E}[\varepsilon_i \mid Z_i] = 0$, which implies the moment conditions $\mathbb{E}[Z_i\, \varepsilon_i] = 0$. The constant imposes $\mathbb{E}[\varepsilon_i] = 0$, and each indicator imposes orthogonality cell by cell.

- **Original model:** $Z_i = (1,\ \text{elite}_i,\ \text{leftgp}_i,\ \text{interact}_i)'$ gives four moments for the four parameters (b_0, b_1, b_2, b_3).
- **Extended model:** estimated on poor households only ($\Theta_i = 0$), as implied by the propositions in Section (REF). The instruments $Z_i = (1,\ \text{Non-Elite}_i \times \text{Left GP}_i)'$ give two moments for the two parameters (b_0, b_1). For rich households ($\Theta_i = 1$), (**??**) reduces to $\tilde{y}_i = \varepsilon_i$ and does not depend on b. two moments for the two parameters (b_0, b_1).

Both models are exactly identified, so the two-step `GMM` estimator coincides with the one-step estimator.

B.1.4 Normalisation of clientelistic pressure

The recovered z is mapped onto $(0, 1)$ with a logistic transformation. This allows the two models to be compared and matches the interpretation of clientelistic pressure as a bounded measure of influence.

B.2 Estimation

B.2.1 Residual and moment conditions.

For fixed (Θ_i, γ), the `GMM` residual implied by (21) is

$$\varepsilon_i(b) = \tilde{y}_i - \ln\Big(\Theta_i + (1 - \Theta_i)\,\gamma\,\exp(x_i' b)\Big). \tag{B.1}$$

The instruments are the group indicators plus a constant, $Z_i = x_i$. Group traits are predetermined: they are fixed before ε_i (measurement and implementation noise) is realised and do not respond to it. Hence $\mathbb{E}[\varepsilon_i \mid Z_i] = 0$, which gives the moment conditions

$$\mathbb{E}[Z_i\,\varepsilon_i(b)] = 0. \tag{B.2}$$

The two models specialise (B.1) as follows.

- **First Specification:** $\Theta_i = \theta$ and $x_i = (1,\ \text{elite}_i,\ \text{leftgp}_i,\ \text{interact}_i)'$. Four moments identify the four parameters (b_0, b_1, b_2, b_3).
- **Second Specification:** estimated on poor households only, so $\Theta_i = 0$ and the residual is linear in b:

$$\varepsilon_i(b) = \tilde{y}_i - \ln\gamma - x_i' b, \qquad x_i = (1,\ \text{Non-Elite}_i \times \text{Left GP}_i)'.$$

  Two moments identify the two parameters (b_0, b_1).

Both models are exactly identified.

B.2.2 Two-step `GMM`.

Let $\bar{g}(b) = \frac{1}{N}\sum_i Z_i\,\varepsilon_i(b)$ denote the sample moments. The estimator is

$$\hat{b} = \arg\min_b\ \bar{g}(b)'\,W\,\bar{g}(b). \tag{B.3}$$

In the first step $W = I$, giving $\hat{b}^{(1)}$. In the second step $W = \hat{S}^{-1}$, where $\hat{S}$ is the cluster-robust estimate computed from the first-step residuals $\hat{\varepsilon}_i = \varepsilon_i(\hat{b}^{(1)})$:

$$\hat{S} = \frac{1}{N}\sum_c \Big(\sum_{i\in c} Z_i\,\hat{\varepsilon}_i\Big)\Big(\sum_{i\in c} Z_i\,\hat{\varepsilon}_i\Big)', \tag{B.4}$$

and c indexes villages. Because both models are exactly identified, $\bar{g}(\hat{b}) = 0$ for any W. The two steps therefore yield the same point estimates; the weighting matrix only affects the variance.

B.2.3 Cluster-robust variance.

Let $m_i = \exp(x_i'b)$ and $w_i = (1-\Theta_i)\gamma m_i/(\Theta_i + (1-\Theta_i)\gamma m_i)$. The derivative of the residual is $\partial\varepsilon_i/\partial b = -w_i\, x_i$, where $w_i = 1$ in the extended model. The Jacobian of the sample moments is $G = \frac{1}{N}\sum_i Z_i\, \partial\varepsilon_i/\partial b'$, and the variance of $\hat{b}$ is

$$\hat{V} = \frac{1}{N}\,(G'WG)^{-1}(G'W\hat{S}WG)(G'WG)^{-1} \times \frac{M}{M-1}, \tag{B.5}$$

where $W = \hat{S}^{-1}$, $\hat{S}$ is recomputed at $\hat{b}$, and $M/(M-1)$ is the small-sample adjustment for M clusters.

B.2.4 Clientelistic pressure and standard errors.

Clientelistic pressure for group (g, r) is the logistic transform of the corresponding linear index,

$$z_{g,r} = \Lambda(c_{g,r}'\hat{b}), \qquad \Lambda(a) = \frac{e^a}{1+e^a}, \tag{B.6}$$

which maps z onto $(0, 1)$ and makes the two models comparable. The selection vectors $c_{g,r}$ are listed below. Since $\Lambda'(a) = \Lambda(a)[1-\Lambda(a)]$, the delta method gives

$$\widehat{\mathrm{se}}(z_{g,r}) = z_{g,r}\,(1-z_{g,r})\,\sqrt{c_{g,r}'\,\hat{V}\,c_{g,r}}. \tag{B.7}$$

- **First Specification:** $c_{NE,NL} = (1,0,0,0)'$, $c_{E,NL} = (1,1,0,0)'$, $c_{NE,L} = (1,0,1,0)'$, $c_{E,L} = (1,1,1,1)'$.
- **Second Specification:** $c_{NE,NL} = (1,0)'$, $c_{NE,L} = (1,1)'$.

B.2.5 Sensitivity analysis.

In the first model, $\theta = 0.515$ is fixed and re-estimate (B.3)–(B.7) on a grid $\gamma \in (0.5, 1)$. We then repeat the exercise on a mesh $(\gamma, \theta) \in (0.5, 1) \times (0.01, 0.99)$.

In the second specification, Θ_i is determined by the theory, so only γ needs to be varied. Because the residual is linear, $\ln\gamma$ enters only through the intercept: $\hat{b}_1$ and $\hat{V}$ do not depend on γ, and $\hat{b}_0(\gamma) = \hat{b}_0(1) - \ln\gamma$. The grid over $\gamma \in (0.5, 1)$ therefore traces out a known shift in the intercept. As γ rises, $\hat{b}_0$ falls, and both $z_{NE,NL}$ and $z_{NE,L}$ decline.

B.3 Equilibrium Win Probability: Details

B.3.1 The zero-transfer restriction.

The vote-share function in Bardhan and Mookherjee (2023) depends on the difference between the transfers offered by the two parties. The data record only the transfers delivered by the winner, $q^P_{g,r}$, and contain no information on the loser's offer, $q^{-P}_{g,r}$. We therefore set $q^{-P}_{g,r} = 0$.

For the clientelistic component, this restriction follows from the timing of the model. Clientelistic benefits can only be delivered by the party in office, so once we condition on the realised winner, the loser's clientelistic term,

$$(1-\gamma)\, z^{-P}_{g,r}[u(\omega_{g,r} + q^{-P}_{g,r}) - u(\omega_{g,r})],$$

vanishes. For the programmatic component, the restriction replaces the loser's unobserved transfer with the household's endowment $\omega_{g,r}$. The specification then measures the gain from the observed transfer relative to no transfer, turning the original differential (winner versus loser) formulation into an absolute (winner versus no transfer) one.

Under this restriction, instrumental and non-instrumental voters evaluate the same utility gain $\Delta u_{g,r}$. This term factors out of the vote-share expression, leaving a convex combination of the programmatic response, weighted by θ, and the clientelistic response, weighted by $(1-\theta)\gamma z_{g,r}(\gamma)$. We allow z to vary with γ because a party's ability to deliver clientelistic benefits depends on its probability of holding office.

B.3.2 Aggregation to groups.

Transfers and endowments are observed at the group level. We therefore replace each individual's swing density σ_i with the group average $\bar{\sigma}_{g,r}$, and each individual's population weight μ_i with the group share $\mu_{g,r}$. The clientelistic pressures $z_{g,r}$ are then group-level structural parameters.

B.3.3 Second Specification.

Because Θ varies by household type within a group, we aggregate separately over rich and poor households in each group (g, r), using their population shares $\mu_{g,r}$, their average swing propensities $\bar{\sigma}_{g,r}$, and their utility gains $\Delta u_{g,r}$. Rich households contribute only a constant programmatic term to (25), so only the poor households' term depends on γ. Since the extended

model's intercept shifts exactly with $\ln\gamma$ (Appendix: B.2.1), $z_{g,r}(\gamma)$ is available in closed form,

$$z_{g,r}(\gamma) = \Lambda(c'_{g,r}\hat{b}(1) - \ln\gamma),$$

B.3.4 Algorithm.

For each region, we start from $\gamma_r^{(0)} = 0.75$ and iterate

$$\gamma_r^{(n+1)} = \phi(\frac{1}{2} + A_r + B_r(\gamma_r^{(n)})),$$

until $|\gamma_r^{(n+1)} - \gamma_r^{(n)}| < 10^{-10}$. At each iterate, $z_{g,r}(\gamma)$ is obtained by linear interpolation between the `GMM` estimates on the grid $\gamma \in (0.5, 1)$. If an iterate falls outside the grid, $z_{g,r}$ is held at its value at the nearest grid endpoint.

B.3.5 Convergence.

Write $T_r(\gamma) = \phi(\frac{1}{2} + A_r + B_r(\gamma))$. Since $\phi'(s) = \kappa\,\phi(s)\,[1 - \phi(s)] \leq \kappa/4$,

$$|T'_r(\gamma)| \leq \frac{\kappa}{4}\,|B'_r(\gamma)|, \qquad B'_r(\gamma) = (1-\theta)\sum_g \mu_{g,r}\,\bar{\sigma}_{g,r}\,\Delta u_{g,r}[z_{g,r}(\gamma) + \gamma\, z'_{g,r}(\gamma)].$$

T_r is therefore a contraction, with a unique fixed point to which the iteration converges, whenever $\frac{\kappa}{4}\sup_\gamma|B'_r(\gamma)| < 1$. Because $z_{g,r}(\gamma)$ is piecewise linear, B'_r exists on each grid interval and the supremum is taken over them. In the data this bound is well below one in both regions.

B.3.6 Sensitivity to θ.

Holding $\kappa = 8$, we re-solve (24) for each θ on a grid over $(0.01, 0.99)$, re-estimating $z_{g,r}(\gamma)$ at each θ, and report $\gamma_r^*(\theta)$.

B.4 Mapping the theory to the data

The structural objects in section (21) map to the variables in the data file as follows. The estimation sample excludes the village `RAUTARA`, and standard errors are clustered at the level of `group`.

Theoretical object	Data variable	Definition / coding	Source
$\tilde{y}_i$ (transformed outcome)	`y_tilde`	continuous	`data`
elite_i (elite indicator)	`elite`	1 = Elite, 0 = Non-Elite Elite are defined as cultivator and landowner	`data`
leftgp_i (left-GP indicator)	`leftgp2008`	1 = Left, 0 = Non-Left	`data`
interact_i (interaction)	`interact`	$\text{elite}_i \times \text{leftgp}_i \in \{0, 1\}$	`data`
cluster unit	`village`	clustering for SEs (88 villages)	`data`
excluded unit	`village`	drop `village = "RAUTARA"`	`data`
θ (share parameter)	—	fixed / grid $\in (0.01, 0.99)$	`exogenous`
γ (scaling parameter)	—	fixed / grid $\in (0.5, 1.0)$	`calibrated`
$\bar{\sigma}_{gr}$	—	swing parameter per group.	`data`

Table B.1: Mapping of theoretical objects to data variables

B.5 Measuring Swing Factor

Swing is measured through the straw polls. We know which party the households voted for in 2004 straw poll and 2011 straw poll.

Recency Adjusted Voter Swing

From the survey data, we use two variables which indicate if the household voted for incumbent in 2004 and in 2011.

$$v_{2004} \in \{0, 1\} \quad \text{Indicator hh voted for the incumbent in 2004}$$
$$v_{2011} \in \{0, 1\} \quad \text{Indicator hh voted for the incumbent in 2011}$$

The swing index assigns values $[0, 1]$.

$$\sigma_i = 1 - \left(\frac{1}{3}\, v_{i,2004} + \frac{2}{3}\, v_{i,2011} \right) \quad ; i = \text{households}$$

The new swing index captures an individual's propensity to vote against the incumbent, using their voting behaviour across two electoral cycle (2004 and 2011) to construct a continuous measure ranging from 0 to 1.

The construction of the swing measure is based on voting behaviour across two electoral periods. A voter who consistently opposed the incumbent in both elections is classified as 'swing' (anti-incumbent) voter, while the one who consistently supported the incumbent is considered 'loyal'. Voters who exhibit mixed behaviour across the two periods are assigned intermediate values,

reflecting partial responsiveness.

Importantly, the measure incorporates a recency-weighting scheme. Voting behaviour in 2011 is treated as more informative of underlying swing propensity than behaviour in 2004, based on the assumption that more recent actions capture current preferences. This approach is consistent with the idea of recency bias, where recent experiences have a stronger influence on decision making.

Formally, the index can be interpreted as a recency -weighted measure of incumbent loyalty, which is then transformed to reflect anti-incumbent behaviour. The weights such as 1/3 and 2/3 assign greater importance to the 2011 vote relative to the 2004 vote, ensuring that recent electoral behaviour has a larger impact on the constructed measure.

Using these household-level swing values, we compute average swing measure across groups defined by political alignment and initial endowments. This aggregation allows us to link voter responsiveness to the key dimensions of the model.

$$\bar{\sigma}_{g,r} = \frac{1}{N_{g,r}} \cdot \sum_{i \in G(g,r)} \sigma_i$$

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